Thesis Proposal

Understanding Artificial Intelligence as a Material for Design

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ABSTRACT Technology often comes into being in the form of a technical advance, and without much concern for user experience (UX) design. As the technology matures, designers work to re-understand how a technical advance might be situated in the real world and to invent new product forms not imagined when the technology was first invented in an engineering lab. Design innovation is the work designers do to reframe the purpose of a technical advance and to generate novel forms that lead to new products and services.

Technical advances of AI increasingly shape how users interact with products and services. However, it is difficult to see design innovations taking place with AI. It is hard to trace how an AI technical advance produces many new product forms that operationalize the tech for new target users, new activities, and/or new contexts. When working with AI, designers reported challenges in almost all archetypal design activities (e.g., opportunity finding, sketching, fast and iterative experience prototyping, reflection and identification of unintended design consequences), on many familiar design issues (e.g., ethics and trust), spanning a wide variety of subject domains.

My thesis work focuses on understanding AI as a *design material*. I ask: *What exactly makes AI distinctive, and why is it so difficult to design? How can HCI and design research make it easier to understand and manipulate into new products and services or into the innovation of existing products and services?*

I began by synthesising the existing uses of AI listed in HCI research, identifying pathways between the technical advances and the UX value they offer. I then investigated how designers in the industry who have the most experience working with AI have innovated in their practices. Reflecting on these two exploratory studies, I identified two key characteristics of AI that shapes how it is designed and experienced: multi-layered capabilities and the decoupling between capabilities and system behaviors. I argue that these two characteristics together form a useful framework and provide structure for more effectively describing AI's design space.

Using the framework as a scaffold, I describe a set of Research through Design projects (two completed and one proposed) and discuss how the tools and knowledge emerging from these concrete design situations contribute to the broader challenges of innovating AI products and services. These projects include (1) designing a prognostic decision-support system that helps clinicians decide whether and when to implant an artificial heart to a heart-failure patient; (2) designing an intelligent text editor; (3) supporting the fan fiction writing-and-consumption ecology with language intelligence.

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1 INTRODUCTION

Technology often comes into being in the form of a technical advance, and without much concern for design. As the technology matures, designers work to re-understand it and to invent new forms not imagined when the technology was first invented. Mobile computing offers an example of the transfer from technical advance to user experience (UX) innovation. The mobile phone was invented as a communication device that makes voice calls and send text messages; Design thinking advanced this technical innovation by considering it more as mobile computers and less as telephones. This shift brought about phones on which people can play the Snake game, view calendar, keep address book, set alarm clock, and play music. As the Internet reached mobile phones, design innovation then brought about the combined experience of downloadable wallpapers, ringtones, games and applications, further recasting mobile phone as a personalized possession and a gateway to individually curated services. Such services include phone calls over wifi, that challenge the idea of owning a "phone" at all. Designers are now helping create new uses of mobile phone as hubs of sensors and outlets of Internet of Things. These new service designs situate mobile devices valuably in people's day-to-day lives and contexts, creating environments saturated with computing. In each of these cases, a technical advance triggered an wave for design innovation to generate a wealth of new product forms that operationalize the technology for new users, new activities, and new use contexts. Such design innovations in turn serve as new platforms for further technical advances.

Recently, some HCI researchers have started to discuss that technical advances of AI are ready for similar design innovation.

1.1 The Vision of AI as Material for UX Design

Technologies generally considered as AI increasingly shaping how users interact with technology products and services, even though, like "mobile computing", the notion of AI is a fuzzy one. From mundane spam filters to personalized newsfeeds on social media, to conversational agents at home like Alexa to the promise of autonomous cars, many products and services are already improving user experience (UX) with algorithms that learn from an underlying data source. New frontiers of AI continue to open up fresh territories for UX design. For example, language intelligence techniques make it possible to computationally comprehend, interpret, and even generate human language, enabling otherwise impossible modes of interactions. Some in the industry even speculated that AI is the new UI [18], and AI algorithms as being "where the action is" [38].

Both UX practitioners and researchers noted this trend and began taking actions toward envisioning new products and services that leverage AI's technical advances. However, they encountered challenges in almost all archetypal design *activities*, on many design *issues*, spanning a wide variety of subject *domains*: At the start, what is AI is itself fuzzy and debatable. Further, some designers struggled to identify actionable ways to bring a human-centered view to bear on AI [30,41,102,103]; some struggled to sketch fluid, adaptive behaviors that AI systems exhibit [108]; and some struggled to prototype AI systems due to their unwieldy technical complexity [30,80]. Others found that even tackling familiar design issues is challenging in the context of AI, such as managing user expectations, and account for the designs' ethics and societal consequences [7,51,73,74]. These challenges, among others, have been reported in various subject domains from healthcare to smart homes, environment, and city [4,5,40], under various banners such as designing data [15,35,71], designing IoT [39,97], designing context-aware interactions [82] and more.

As a result, it is difficult to see such type of design innovation as mobile computing taking place with AI. It is hard to trace an AI technical advance being followed by many new product forms that operationalize the tech for a new target

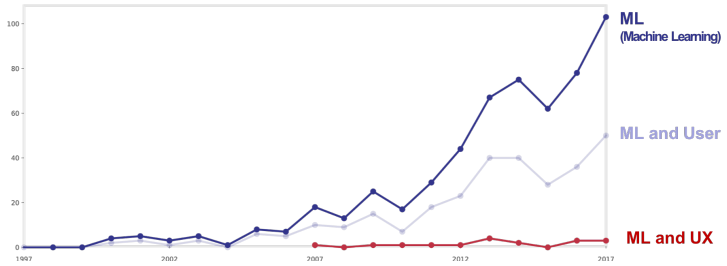

Fig. 1. The number of SIGCHI publications that mention ML and UX in their meta-data

user, a new activity, and/or a new context; the forms beyond what the technical advance initially intended. Even in HCI research, the UX of AI can be seen as lagged behind AI's technical advances.

Why does AI seem to be distinctively difficult to design? UX designers regularly integrate new or partially understood technologies into novel and valuable new products and services.

To envision things that never before existed, designers engage in reflective conversations with their design materials. Let me use cardboard as an overly simplistic example. When designing with cardboard, designers bend, tear, fold it in various ways, observe how it responds and consider what new design possibilities may emerge [84]. Through this "*reflective conversation with the material*", designers explore a wide range of design alternatives. They both identify the right thing to make by ideating many possibilities, and they make the thing right through iterative refinement [20,72].

Such reflective, material conversations are not yet happening in designing AI. A number of symptoms may illustrate this. First, experienced designers currently fail to notice obvious places where AI can improve UX [108]. Second, designers typically join AI projects near the end, after the functional decisions have been made. Designers are not asked to envision what an AI product or service could be, but instead only asked to help situate pre-determined AI functions into a context for a preselected set of users. "*Design teams are simply putting lipstick on the pig*" [30]. Lastly, researchers noted a lack of design innovation in AI systems. "*Designers often have cliched understandings of this medium, driven by the hype or criticism surrounding the field. For example, some typical stereotypes are that wearables are for fitness, AI is for automating tasks.*" [97]

I draw connections between the challenges that HCI has identified in designing AI systems and the challenges designers face in having a conversation the material of AI. In my thesis work, I have synthesized the existing uses of AI in HCI research and investigated how experienced designers in the industry innovated AI in their respective practices. I have also designed AI systems for a range of application domains, spanning from clinical machine learning to language-based interactions. Reflecting on this rich set of experiences, I attempt to articulate what exactly about AI makes it seem more difficult to design than most other familiar technologies and how HCI and design research might go about making it easier.

1.2 Scope and Orientation

1.2.1 Defining Artificial Intelligence (AI)

A investigation of AI as a design material is hampered at the start by the difficulty of pinning down a precise definition of AI, one that competes with its popular portrayal [91]. Even in technical terms, the scope of AI is disputed and a

moving target [1] [1]. More importantly, this thesis focuses on a design practice view of AI, a kind of understanding that is likely to be very different from its technical definition (Figure 2). In fact, the very goal of this thesis is to explore what AI is or means for UX design, what aspects of it are relevant to design, and how designers should meaningfully relate to it.

Therefore, I do not intend to draw an accurate or sophisticated technical boundary of AI here. Instead, I identity a point of departure for these explorations by proposing a broad, operational bounding of AI:

> Of particular interest of this thesis are *computer programs that make inferences based on data.* These inferences are non-deterministic from the perception of a user. This includes but is not limited to classic AI systems, crowd-sourcing systems, and expert systems.

This bounding places only two initial restrictions for my upcoming explorations: First, my investigation is premised upon AI as a kind of technology, rather than a human perception of systems being “smart" or “intelligent" (as in smart or intelligent home/work/car/city/etc.). Secondly, my investigation excludes generic computer programs that execute manually-specified procedures or rules on data, such as JavaScript and web browser. Through the course of my thesis, I take this bounding as a point of departure, examining and amending it as I search for better interpretations of AI that are more specific and useful to UX design.

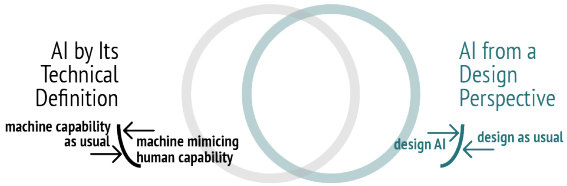

Fig. 2. Technical vs. design definition of AI

1.2.2 AI as Design Material

The idea of AI as a design material roots in the theoretical distinction between designing technology at an application level and at a material level [72,76]. For the former, designers give forms to the technology’s given functionality; For the latter, designers engage with the technology’s capabilities and potentials and actively explore what new aesthetics, functionality or experiences it may afford [76,99].

Similarly, the idea of AI as a design material promotes a vision in which designers actively engage with AI’s capabilities and potentials, and envision a wealth of new, valuable uses that are unlikely to come out of a technical AI research lab (Figure 3). This proposition asserts a series of not-yet-answered questions about AI’s material properties:

- How may designers understand AI from a design perspective (as they understand cardboard via notions of weight and strength)?
- How may designers go about “having a conversation" with AI in their respective design situations (as they bend, tear, and fold cardboard)?

[1]The term artificial intelligence generally refers to machine capabilities that mimic the efficiency and robustness by which the human brain represents information. Modern machine capabilities that are generally considered as AI mostly learn such representations from large amounts of data, therefore the term AI is sometimes used interchangeably with “machine learning" (ML). As machines become even more capable, new frontiers of AI are likely to continue to evolve [81].

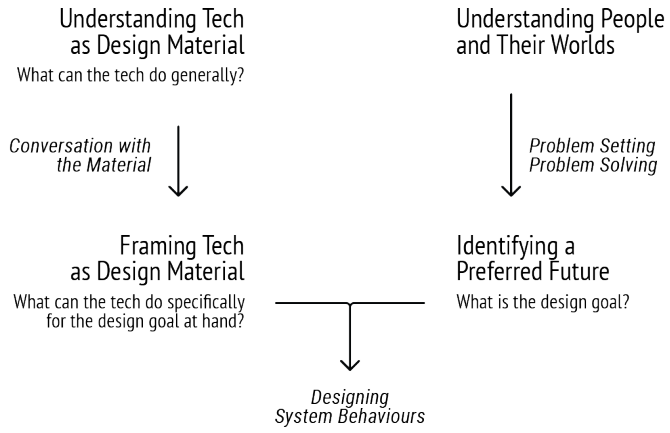

Fig. 3. How design material fits into design processes. We draw attention to the distinction between designers' felt understanding of a design material (1) and their situated conversations with the material (2).

The material lens of AI offers an alternative sensibility to the challenges of designing AI systems. HCI research has previously reported such challenges on every step of a typical design process. This body of work typically focused on the ultimate particular designs of AI and how they manifest in the work (Figure 3 right side). In contrast, few studies have investigated these challenges through the lens of designers' conversation with the technology material (left). For example, researchers have rarely tied unexpected UX breakdowns of AI systems to the challenges of iteratively prototyping and testing them.

1.3 Research Approach

The overarching goal of this thesis is to better understand the challenges of working with AI as a design material and to help UX researchers and practitioners to more easily work with it.

This is a nascent research area where even the exact challenges of designing AI are not yet fully understood. Two signs may demonstrate this. First, the various challenges of designing AI remains unsorted. The aforementioned challenges across design activities, design issues, and subject domains are evidently interrelated, but it remains unclear how they relate. There exists no agreed-upon set of themes around which we can summarize the challenges AI imposes. Secondly, many reported challenges of designing AI seemed merely symptoms of some deeper challenges that AI brings. For example, AI systems are widely considered difficult to prototype due to their technical complexity and demand for data [30,57,80,96]. The paradox is that numerous HCI prototyping methods suggest that designers *can* grapple with resource-intensive, complex technologies via inexpensive, simple artifacts (e.g., paper prototypes, Wizard of Oz systems). What sets AI apart from other technically complex systems? What exactly makes AI difficult to prototype? Such discussion is central to the booming research on prototyping AI [57,80,96], yet largely absent.

The lack of systematic and elaborate understanding of AI's design challenges makes it difficult to address them. This can manifest in the work of HCI/design researchers in at least two ways. On one hand, they do not know what design challenges or tasks they should prioritize and study, because they do not know how the challenges are dependent on each other. On the other hand, they cannot easily separate their findings that are specific to the case from those generalizable to other instances of AI. Unable to articulate the fundamental challenges of designing AI, they cannot

easily build upwards toward its overall solutions, by solving a set of smaller problems I encountered in distinctive design situations.

Therefore, I began my thesis work by considering why designing AI appears to be such a challenge in the first place. To do so, I synthesized 2,494 articles from AI-related HCI literature [102], interviewed more than 70 practitioners active in this area from many research institutions and enterprises [103,106], and hosted workshops with students to better understand the challenges novice designers encounter. Theses studies span in a wide variety of AI's subdomains and distinctive design situations, and reveal a suite of possible approaches to understanding AI.

Meanwhile, I took on Research through Design (RtD) projects and designed the UX of a number of AI applications, from the adaptive interfaces of a context-aware mobile app [108], to a clinical decision-support system that helps cardiology teams decide whether and when to implant an artificial heart to a heart failure patient [104,107], to envisioning writing assistance features in a Word document editor [105]. These first-hand experiences became the ground for developing and re-interpreting the emergent framings of AI as a design material.

The two threads of work informed each other and developed simultaneously. In the process of formulating this proposal, I reflect on these intertwining threads of work and attempt to propose a framework for conceptualizing AI as a design material. I have chosen this framework for its usefulness in three respects:

- *Analytical leverage*: It helps scaffold and synthesize a full range of opportunities and challenges of designing AI;
- *Explanatory power*: It helps specify the contribution and limits of emergent design knowledge;
- *Constructive potential*: It provides more value than objectively organizing known design challenges. It offers valuable new insights into ways to addressing the challenges.

Like many conceptual framework proposals, this framework is not fixed nor final. I plan to complete one more RtD project on designing artificial language intelligence, specifically, designing for fanfiction writers with artificial language intelligence for the fanfiction writing-and-consumption ecology. I will also continue to examine and improve the framework as I continue my RtD work and as my thesis work continues to develop.

1.4 Proposal Overview

This proposal is organized to progress from conceptual questions such as "what makes AI seem to be uniquely difficult to design" to the concrete, first-person experiences of designing AI applications.

In Chapter 2, I provide an overview of the challenges of designing AI and other difficult technologies, and how HCI has been addressing these challenges. This overview shows how AI has introduced new challenges and started to destabilize these conventions. It also suggests that existing research about UX design of AI has so far been focusing on making its interactions right. My work advances on this by bringing in the question of how can we widely explore many design possibilities to identify the right things to make.

Next, I describe two of exploratory studies that narrowly focused on these aspects of designing AI. One study is an HCI literature synthesis through the lens of "match-making", in which I identified concrete handles for AI's capabilities and design innovations, as well as stepping stones that bridge the two. The other study investigated how seasoned UX designers in the industry understand and work with AI. The findings of these studies highlighted a small set of pragmatic questions designers grapple with in practice, for example:

- How may designers conceptualize AI's capabilities?
- How may designers sketch or fast-prototype AI-mediated interactions?

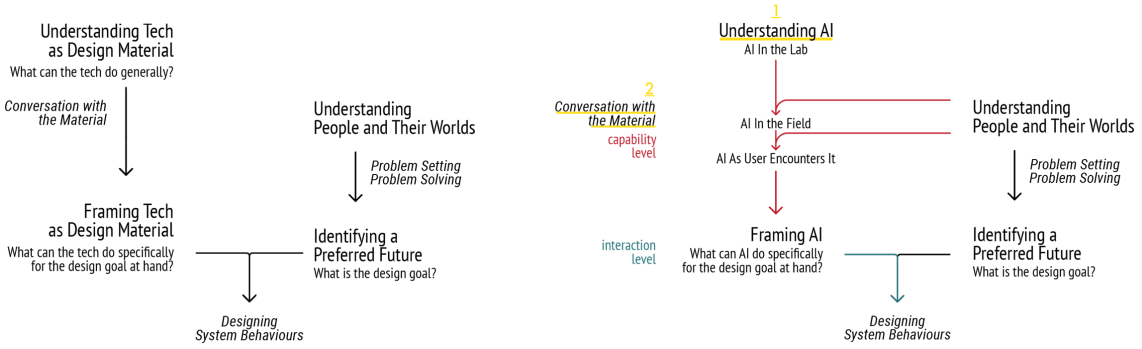

Fig. 4. Contrasting AI (right) vs. other technology materials (left)

- How may designers effectively communicate the design ideas underlying AI interactions to users, other designers and collaborating AI engineers?
- Do designers need technical AI knowledge to design? If so, in what aspect and to what extent?

Reflecting on both related work and my explorations, I propose one framework for conceptualizing AI as a design material. Structuring the framework are two distinctive yet interrelated characteristics of AI: One is its multi-layered capabilities, that AI can possess distinctive capabilities in the lab, in the field, and when being encountered by individual users; The other is the decoupling of AI capabilities and its system behaviors. I describe how these two characteristics of AI can together serve as an useful conceptual tool for outlining the range of AI's design opportunities as well as for getting a grip of its unique complexities and challenges (Figure 4).

In chapter 5, I describe a set of RtD projects including two completed projects and one remaining. These projects provide first-person accounts of "designerly conversations" with AI when designing AI for healthcare and when designing NLP for writing. I draw on these experiences to reflect on the broader challenges of and solutions to designing AI, and to examine and refine the proposed framework.

Finally, I summarize the lessons learned so far across the exploratory and hand-on design projects, and describe the remaining research questions that I plan to explore in my proposed, fanfic RtD project.

1.5 Expected Contributions

There are three expected contributions of this research. The first contribution is the proposed framework as a scaffold of the opportunity and problem space of improving UX with AI. This scaffold has both generative and evaluative implications. On the generative side, the framework raises the ceiling in terms of what roles UX design may play in advancing AI and unfolds a full range of opportunities in this space for both HCI practitioners and researchers. On the evaluative front, the RtD projects and reflections demonstrate that the framework can help designers and researchers to easily assess the limits of emergent design knowledge.

The second contribution is the set of constructs that connect the conceptual, material questions of AI to the concrete, pragmatic challenges of designing AI. For example, the question "do designers need technical AI knowledge to design" both connects to the theoretical inquiry of what AI *means* for design and has implications for the creation of sketching and prototyping tools. Such conceptual constructs offer valuable stepping stones for constructive design researchers,

enabling them to make a research contribution on AI as a design material while designing at the level of the ultimate particular.

The third contribution is a set of new tools and knowledge that UX practitioners can directly apply to the design of AI systems. Emergent from the RtD projects, these tools include a set of design patterns of machine-learning-ready adaptive user interface wireframes, a set of wireframes that illustrate abstract language interaction design ideas, a new Wizard-of-Oz prototyping method that simulates the likely errors of not-yet-existent deep learning systems, and more.

♣ ♣ ♣

2 RELATED WORK

In this chapter, I map related HCI research on AI and other new or difficult design materials onto the proposed framework. This overview shows how AI has introduced new challenges and started to destabilize these conventions. It also suggests that existing research about UX design of AI has so far been focusing on making its interactions right, and less on the process of exploring many design possibilities to identify the right things to make.

2.1 Challenges of and Supports for Designing New Technology

To envision things that never existed, designers engage in the reflective conversation with the materials at hand. I have previously described this creative process (section 2.1). This creative process works well when designers have a tacit understanding of the materials they are using, of the materials' material properties – what UX capabilities and limits they possess, what new aesthetics, functionality or experiences they may afford [76,99].

However, this creative process happens less readily with new technology, because designers lack a ready-at-hand knowledge of what it is and what it can do. In previous cases, HCI researchers work to transfer the new technologies to practice by creating demonstrational artifacts, toolkits, design patterns, and more. Below I elaborate on two of these approaches that focused on enabling designers' conversation with technology materials: creating artifacts that hold design knowledge, and providing supports for sketching and prototyping.

2.1.1 Artifacts that Hold Knowledge

HCI and design research have routinely created artifacts that communicate the design qualities of a new technology as a research contribution. Research through Design (RtD) researchers made the case that RtD can embed the HCI technical advances into concrete "sensitizing" artifacts, artifacts that would be easier for practitioners to understand, and would sensitize other designers to the design possibilities beyond the specific artifact that was created [110,112]. For example, Moussette was interested in playing with haptics as a design material [66]. He started an ambitious project to sketch with haptics and produced a set of simple devices with very different kinds of haptic feedback embedded. These devices sensitize designers to the range of possible tactile sensations haptics possess. Similarly, HCI researchers have developed sensitizing concepts for interactive textiles, producing new opportunities for the textile industry based on IT innovation [77].

Underlying this body of work is the idea that artifacts hold and codify design knowledge. Very early HCI work proposed that artifacts hold theories that have yet to be discovered [21,28]. Many design researchers have extended this idea of "designerly" ways of knowing, that designers understand design, materials, and knowledge via the designed things. Further, codifying knowledge in artifacts works much better when design researchers repeatedly foray into making around the same or similar set of concerns, instead of proceeding one case at a time [17,60]. As the haptics example has shown, a suite of sensitizing artifacts could become a new language for designers to consider and speak more specifically and systematically about the design qualities of the technology. Others explored ways of constructing intermediate-level, abstracted knowledge from curated sets of particular design instances [52].

2.1.2 Supports for Sketching and Prototyping

Sketching and prototyping are two archetypal activities through which designers' conversation with the material can unfold [20,63]. To enable such conversations with new technologies, research has created new methods and tools for sketching and prototyping. For example, to help get a grip on the design of taste experiences, Obrist et al. created sketches representing the temporal and affective characteristics of taste [70]. Other well-established examples include paper prototypes, experience prototypes, Wizard of Oz systems, technology probes and more.

The iterative process of sketching and prototyping encompasses learning, generative and evaluative design activities. Goldschmidt characterises sketching as a dialectic between seeing-as and seeing-that, where "seeing that" is reflective criticism and "seeing as" is the reinterpretation of the sketch that provokes creativity [43]. Lim et al. highlights the similar dimensions of prototypes: prototypes as filters and prototypes as manifestations of design ideas [63]. In this sense, effective supports for sketching and prototyping could enable designers to 1) develop their respective, contextualized understanding of the material, 2) to explore more design alternatives, 3) to critically assess their designs against the design goals, and 4) to identify unintended consequences [43,63].

2.2 Challenges of and Supports for Designing AI

HCI and design researchers recently noted that there may be a gap between HCI advances that use AI and the design innovation that operationalize these technical advances into many new products and services [4,5,30,40]. This body of work has revealed many unsorted challenges designers face when working with AI. They speculated that "this might be because it (AI) is a new and difficult design material".

2.2.1 Understanding AI

In a 2017 survey, practitioners stated they could not easily envision novel and interesting uses of AI. While they understood how AI worked broadly, they did not understand what was needed to design with it [30]. Based on observations of these challenges, design researchers and practitioners have begun making efforts to help UX designers work with AI as a design material. These efforts advanced in a variety of directions, but they typically fall into two broad characterizations.

One characterization presumes that designers cannot work with a material they do not fully understand, redirecting designers' focus away from the interactive artifact towards the statistical processes of AI products [23,94]. Several books, courses and online collections of materials help designers learn about AI (i.e. [22,42,48,49]). The book *Machine Learning for Designers*, for example, reviews concepts of supervised and unsupervised learning, as well as common analogies of how AI works [48]. This is quite different from the traditional ways of moving new technology from research to design practice, which assume designers do *not* need to understand the technical specifics of the materials they use to design.

The second characterization, less popular than the first, works to demonstrate divergent ways of designing AI, sensitizing designers to its rich design possibilities. Researchers have explored various themes related to the interaction design challenges AI imposes, such as intelligibility [62], animistic design [65], and design for the peace of mind [41]. An university design course asked students to create a "useless artificial intelligence" product, encouraging students to engage with AI through a making process that probes affordances, capabilities, and limitations of this technology [97]. Instead of teaching designers what AI is, this approach challenges designers to envision applications technologists would likely never imagine.

2.2.2 Conceiving of Design Ideas

A 2017 survey reported that design teams in the industry rarely conceive of an entirely new way to use AI and then take this to a development team to implement. Instead, they typically joined projects towards the end, after the functional decisions had been made. "*Design teams are simply putting lipstick on the pig*" [30]. Noteworthily, these designers do not know how to bring their UX expertise to bear on AI, even when they know how AI works broadly. This can seem unusual, since most design research to date assumes that designers, as long as sensitized to a particular design material, are able to "play with" it and explore new design opportunities. Few HCI research has investigated the challenges of design ideation around AI.

2.2.3 Formalizing and Deliberating Design Ideas

HCI research has rarely discussed how to address these challenges of deliberating and formalizing AI design ideas. While offering creative, user-centered systems, researchers in this area typically describe one design solution, followed by its implementation and subsequent user study evaluation.

An exception is the work on prototyping AI with Wizard-of-Oz (WoZ) systems. Researchers simulated system behaviors with wizard or rule-based simulators so as to rapidly explore many interaction possibilities [25,27,55,90,96]. For example, Suede [58] enabled speech-interface designers to quickly test their conversation scripts with wizards. A 2012 review showed that WoZ has been one of the most popular methods for examining human-robot interaction. In particular, 72.2% of the human-robot verbal interaction studies used WoZ [78]. Unlike the aforementioned, common sketching methods, WoZ does not facilitate designers to experiment a technology's capabilities and limits; Instead, it frees designers from the technical complexities and limitations of AI and facilitate experimentation on interactions.

More work reported challenges when sketching and prototyping AI systems. I identify two themes in these challenges.

First is AI's technical complexity. It requires an unwieldy amount of data and effort to even estimate whether an interaction is technically feasible, or to create a functional prototype [30,108]. This challenges fundamental UX mantras like "*fail fast, fail often*" and "*the last thing that you should do when sketching an interactive system is to write code*" [20]. To address these challenges, some researchers speculated that AI would require a new kind of user-centered design process, processes that start with technology and search for valuable new things to make (namely, "match-making") [14,35]. Unfortunately, this approach is both under-investigated as a design method and underutilized by both design researchers and practitioners.

In this light, there is a rising consensus that the AI systems requires a tighter, more frequent collaboration between UX designers and technologies [4,5,40,67,89]. This can be seen as related to previous HCI research on software as a design material, which noted that designers often need to use developers as a proxy in conversations around the possibilities of software design [72]. Girardin and Lathia's investigation of the designer-data scientists collaboration within their

own organization seems to confirm this speculation [41]. In spite of these few works, I found no work investigating how the collaborations between designers and AI engineers have been or should unfold.

Second, AI-powered UX adapts to different users and contexts, evolves over time, and can make bizarre errors that are hard to anticipate. Even if they know generally how AI works, UX practitioners found it difficult to deliberate on such divergent courses of interaction, or to formalize new experiences with such fluidity [30,108].

Advancing on this challenge, an emergent thread of HCI work created interactive ML systems that allow designers (and non-AI experts in general) to experiment with an algorithm's behavior in real time, such as Wekinator for sensor-based music composition [2], yolo for real-time object detection [75], and [56] for creating time-series based AI system. These tools typically adopted graphic interfaces or sensor-based interactions, eliminating the need for designers to write code. However, these tools uniformly focused on particular application domains and limited the types of input designers can experiment.

2.2.4 Attending to Intended and Unintended Design Consequences

Design researchers noted that AI systems can make performance errors seem bizarre and hard to explain, resulting in potentially dissonant user experiences [30]. This can be exemplified by some well-known AI+ UX breakdowns.

- *Lab capability breakdown*: In 2015, Google Photos' image-recognition algorithm mistakenly categorized two African-Americans as gorillas [3]. Although trained on images of people of all races, the algorithms failed to learn comprehensively, nor did they have the cultural context to understand the risk of returning this as an answer.
- *Field capability breakdown*: In 2016, Tay, a Twitter bot that learned conversational engagement skills from the utterances of Twitter user interactions, was shut down after it tweeted racist slurs [69]. In this case, the algorithm delivered the expected capability, yet the problematic open-world data – racist tweets that Tay learned from after its release – led to the product failure.
- *User-experienced capability breakdown*: Many recommendation systems provide worse recommendations to newer users than old users, because new users have not yet cast a sufficient amount of "likes" for the algorithm to learn from. This cold-start problem creates a sub-optimal on-boarding experience and often causes new users to quit [59]. In this case, algorithms again delivered the expected capability and the open-world data effectively enhanced it, yet the technology can be seen to have failed in a highly localized, user-specific way due to limited local data.

It is typically through iterative prototyping and testing that designers critically assess their designs against the UX goals as well as attend to unintended design consequences such as the above ones [43,63]. However, as described earlier, AI systems are difficult to prototype. Research is sparse on known HCI methods' applicability to and effectiveness in prototyping AI systems.

A related and rapidly growing strand of work proposed guidelines and design principles for the interaction design of AI systems. A CHI'19 Microsoft paper proposed 18 generally applicable design guidelines for human-AI interaction, for example, "make clear what the system can do", "make clear how well the system can do what it can do" [7]. The authors noted that these guidelines were optimized for generalizability across all types of AI-infused systems and observable interface properties, therefore lend themselves well to usability inspection methods such as heuristic evaluation. They also noted that these guidelines might not adequately address all types of AI-infused systems. Similarly, Google has previously released 43 rules of machine learning, which encompass its internal best Practices for ML engineering.

Though with an engineering focus, some rules explicitly addressed HCI considerations (e.g., #23 “You are not a typical end user"; #43 “Your friends tend to be the same across different products. Your interests tend not to be".)[113]

In spite of these two sets of guidelines from the industry, most other guidelines available outlined the challenging issues of designing AI. For example, designing for transparency, designing for unpredictability, designing for evolution, etc. Some guidelines focused on specific AI-related application domains such as IoT[61]; Others on AI in general[51]. It remains unclear how these guidelines may affect design practice, since researchers typically concluded these guidelines from their own work experience and rarely assessed the effectiveness on others’ design practice.

2.3 Overview of Previous Work

HCI and design research has so far offered a rich set of methods and insights for making technological advances available for UX design innovation. These methods and insights (e.g., sensitizing artifacts, sketching and prototyping methods) codify intermediate-level design knowledge about the material and are meant to inspire design innovations beyond the particular artifacts that the researchers created.

Such intermediate-level knowledge is lacking in designing AI. Instead, most HCI research on designing AI has so far focused on ultimate particular interactions (e.g. demonstrational AI systems, AI design principles or guidelines) rather than envisioning AI that improve UX in less obvious ways (e.g. sketching or prototyping methods), on making the interactions right rather than envisioning the right things to make. Further, the works that did envision novel designs of AI typically functioned at a high level, privileging blue-sky design ideas of AI (e.g. unrestricted WoZ studies) rather than grounded, state-of-art AI capabilities in the lab, field or real-world users’ experience.

Now is an opportune time to step back and connect these two differing intellectual commitments. One straightforward approach to synthesize the approaches and learnings underlying existing AI systems and relate them to more abstracted, general design knowledge. Research opportunities also lie in designing new sensitizing artifacts that embody a grounded understanding of AI capabilities. My work in the following chapters serves as first steps in these lines of inquiries.

The challenges that practitioners have reported when working with AI as a design material can serve to focus such inquiries, providing research questions when we synthesize varying design instances and allow for reflection. For example:

- How may designers understand AI capabilities?
- Do designers need technical AI knowledge to design? If so, in what aspect and to what extent?
- How may designers recognize opportunities? What made it difficult to recognize AI design opportunities, even for those who understand how AI works generally?
- How may designers sketch and fast prototype AI systems?
- How may designers evaluate AI designs, especially in attending to AI failures that seem hard to anticipate?
- When and how is collaboration with technical expertise necessary in the design process?
- Should user-centered design change?

♣♣♣

3 EXPLORING AI'S DESIGN AND PROBLEM SPACE

The main focus of my thesis is to help UX practitioners more easily work with AI's technical advances, and to help design research make contributions that open up the space for such design innovation. But where to start? The design space, where AI might improve UX, feels significantly larger than the spaces of previous design inquiries like designing haptics. As a first step, I chose to investigate respectively HCI literature [102] and practice [103] in the industry that used AI. While there have been many AI advances outside of HCI literature and practice, they provide a good enough starting place to reveal the AI topics that HCI has investigated, and they generally has a focus on human needs and desires.

3.1 Study 1: Mapping HCI Research to Bootstrap AI-UX Matchmaking

Recent HCI research noted a lack of design innovation with AI, suggesting a potential gap preventing AI technical advances from moving into commercial products [30]. Design researchers have claimed that they can help with the challenge of transferring technical advances to practice [111].

In addressing the challenge of starting with technology and searching for valuable new things to make, HCI researchers have proposed *matchmaking* [14]. Matchmaking starts by asking designers to detail the technical *capabilities* of the tech they are working on. Next, they systematically work to discover *activities* related to these capabilities, *domains* related to the activities, and finally target *users* connected to the revealed domains.

The goal of this study was to reveal the topics HCI has worked on with respect to AI and to identify similar handles for matchmaking. Specifically, We searched for HCI publications that mentioned AI or related keywords in their meta data, curated a corpus of 2,494 publications, and carried our analytic literature review in three phrases:

(1) Revealing the technical topics HCI has extensively investigated to help designers and researchers better grasp the capabilities of AI;
(2) Revealing the UX values that AI has previously been used to create for end users;
(3) Revealing pathways that connect the two;

Collectively, they provided concrete handles for designers and researchers to start "matchmaking" and ideate many new ways of leveraging AI capabilities to improve UX.

3.1.1 Identifying Technical Capability Handles

Our analysis first revealed seven clusters of AI technical capabilities. We labeled them as: 1) intelligent UI and usability; 2) intelligent environment; 3) recommenders and user modeling; 4) social network and sensor framework; 5) AI and knowledge system; 6) search and deep learning) and 7) sentiment analysis and affective computing. The clusters were not necessarily mutually exclusive. Some papers fit in more than one clusters as these publications may speak to multiple literature or multiple application domains. Figure 5 details the topics of and the semantic distances among each of these clusters.

In four of the clusters, there was a noticeable funnel effect, with researchers frequently devising vastly differing AI techniques and coming to similar, convenient interaction forms. A typical example of this can be seen in the Intelligent Environment cluster. Here a wide variety of technical topics - from internet of things to smart homes, classrooms, workplaces and cities - were tied to only a few interaction forms: automation, persuasion or multimodal interfaces. These common pairings illustrate that the application domain has frequently produce these same interaction forms. The vast majority of papers in our corpus seemed to use a "convenient" interaction choice. The different application domains often used different "convenient" design choices.

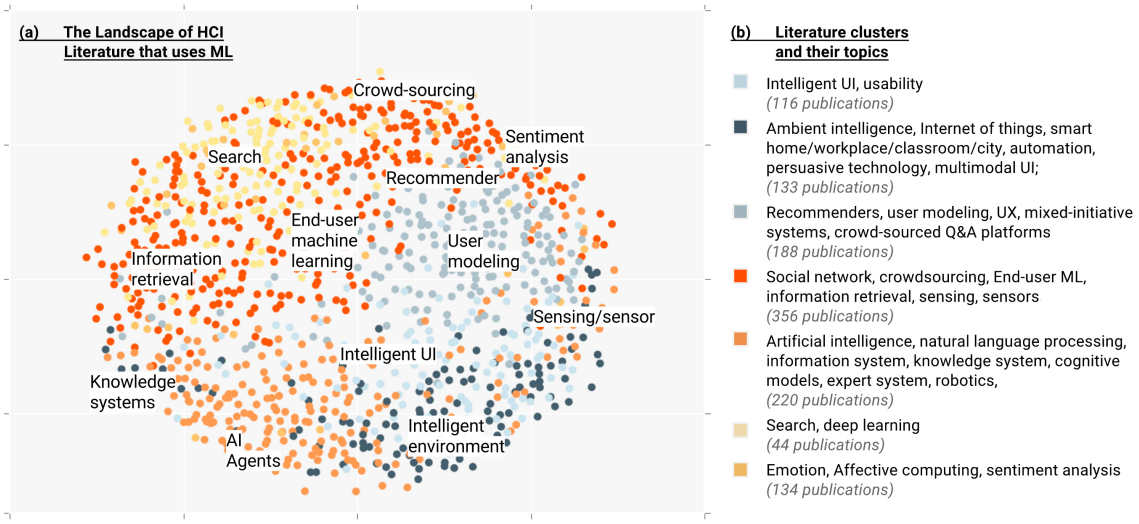

Fig. 5. An overview of HCI research that uses AI or machine learning. (a) An illustration of literature landscape based on the semantic distances among each publication. Each dot represents a publication, color-coded by cluster. (b) major topics of each cluster. The illustration and topic keywords are algorithmically generated; we then manually interpret and annotate the topics.

Topic co-occurrences in these clusters surfaced some common combinations of AI techniques and interaction forms. For example, user modeling appeared often with recommender, suggesting that the latter is a typical use of user-modeling techniques. Likewise, in the sensor network cluster, mobile sensing appeared often with notification, visualizations, and persuasive technologies. Deep learning appeared often with search and not with other interaction-related topics.

The other three clusters center in AI technologies and appear less bound to any particular interaction form. Relatively few papers in relation to these AI technologies have attempted to address the interactive forms they might take. For example, one cluster shows that deep learning falls close to search, and not close to any other topics. This suggests that research on deep learning within HCI might predominantly have been used to improve the search user experience.

3.1.2 Identifying UX Value Handles

The clusters above provide a set of technically defined starting places to AI's design innovations. However, HCI research and practice employs a user-centered orientation that discourages starting design work from the perspective of a specific technology and then searching for users. To aid in making these seven clusters more approachable from a user-centered process, next, we focused on how AI has previously been used to create new value for end users.

Our Analysis shows four channels where AI creates or augments value for users. This could help designers generate many concepts from the perspectives of different possible stakeholders.

- inferences about self (e.g., automatically inferring and logging sleep stage and sleep quality [45])
- inferences about the world (e.g. prediction of public health crisis outbreak [46])
- inferences about what might be optimal (e.g. recognizing the patterns of engaging conversations to design engagement-aware conversational agents [101];)

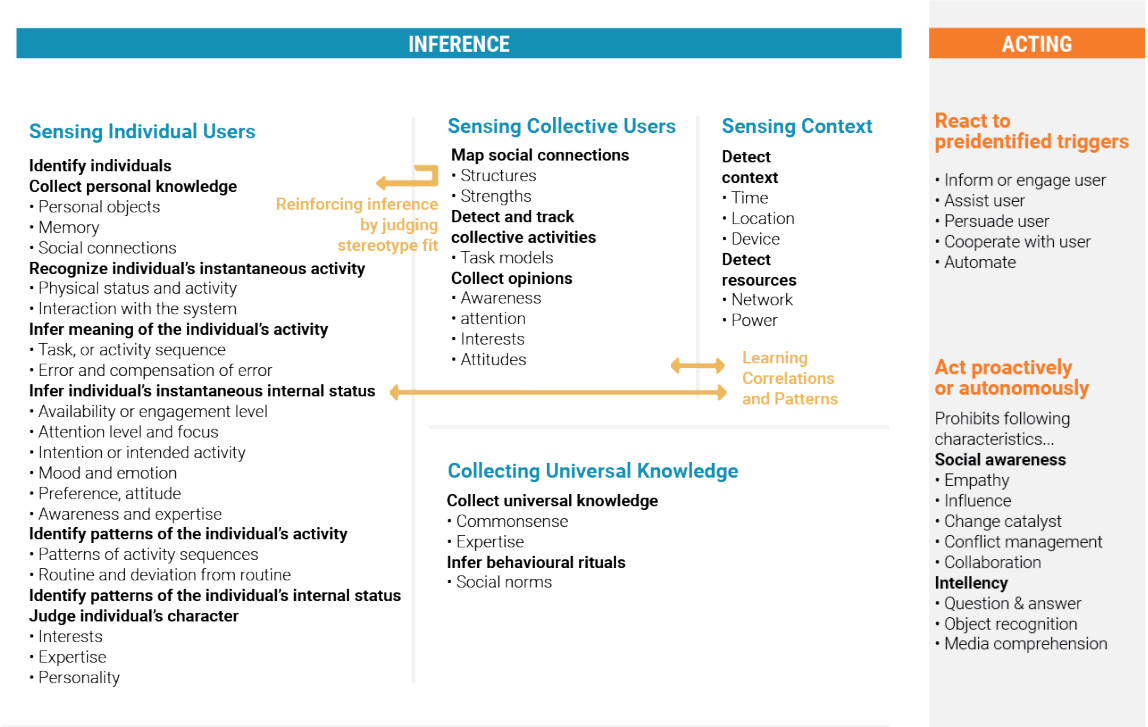

Fig. 6. A schema of AI/ML capabilities in terms of enhancing inference and actuation. These capabilities increase a user's perception of the aforementioned experiential value.

- inferences that provide utility and/or a new capability that is not directly related to self-understanding, contextual awareness, or optimization. Sort of a catch-all for everything else. (e.g. the ability to turn people's skin into a touch interface because devices might shrink to be too small to effectively interact with [47]).

3.1.3 Identifying Interaction Pathways that Connect Capability and UX Value

Finally, we created Figure 6 to illustrate the pathways through which AI capabilities HCI researchers have developed can connect to concrete interactions and users' experiential values.

3.1.4 Summary

This literature study provided an analysis of the current landscape of HCI research in relation to AI, and organized the diverse design opportunities that comprise this field. The two constructs (technical topics and user value channels) and pathways connecting the two provided a concrete way for designers and researchers to begin to investigate AI in their own work. These constructs took stock of what clusters of technical capabilities are available; They are not the end, but instead they are meant as a starting point for an iterative user-centered design process through which the designs will both evolve and be reframed.

3.2 Study 2: Investigating How Experienced Designers Worked with AI

Previous research had identified many problems designers face when working with AI or working on tools meant to make AI more accessible to designers. With these challenges in mind, we investigated the design practices of some of the few UX designers who regularly create new products and services that use AI to enhance UX. Contrary to the holistic and analytical literature study, this study gives prominence to designers' subject handling of AI when they design. We hoped that their approach and reflections would reveal new insights on the kinds of tools or knowledge needed for enhancing UX with AI.

We interviewed 13 designers who all had at least four years of experience designing AI enhanced UX; half of them had more than 10 years of experience. The interviews produced several interesting findings: 1) Designers shared that they knew very little about how AI works, and this was not a priority for them. They instead used designerly abstractions and popular exemplars to explain what AI is and to communicate design ideas with each other. 2) Designers "play" with quantitative data during all phases of a design project. 3) AI projects are longer in preparation and scope than other design projects. During the preparation stage, designers evolved their ideas in close collaboration with data scientists; They did not deliver fully formed designs to a technical team.

Noteworthily, we noticed two distinct work cultures that had a large impact on how participants worked. A few participants worked in large, AI-focused organizations, where designers had access to many data scientists and the companies actively promoted innovation via AI. The others worked in environments with much more limited access to data scientists. In the detailed findings below, we note where this difference impacted the work they described.

3.2.1 How Experienced Designers Understand AI

Participants almost all stated that they "*know very little*" about how AI works. They characterized their AI literacy as "*understanding at a very high level... [at the level of] knowing what a classifier is and what a label is.*" Only one participant (P6) had taken any AI course; an online course taken well after graduation. P6 was the only participant with a CS degree, and had transitioned from CS to UX while earning an MHCI degree. All other participants claimed they gained familiarity with AI mostly by reading online articles found on Medium and Reddit.

Interestingly, participants did not feel their lack of technical knowledge hindered their ability to design or to collaborate with data scientists. They shared that they worked on the design issues, not the technical issues, and that working with AI required "*âĂę more design savvy*". Several participants claimed designing the user interaction was "*the actual challenge.*"

> *I don't think of ML (machine learning) as affecting my interaction designs or not. I think about it more like impacting certain algorithms that are inputs into users' experience.* (P14)

This issue often came up when they made illustrations of their AI systems. Many drew a line in the middle, creating a distinct break between the users and the technology. P10's illustration, for example, had a dense collection of design factors on the left: usage scenario, user's frustration, and "*a very simple thing*" the design team thought AI could do to aid the user. They then drew a line followed by several boxes representing the back-end mechanisms. The AI was one of these boxes, and they noted it was really not so simple.

Most claimed that they learned to work with AI similarly to how they learned to work with other interactive technologies. "*Machine learning is just like JavaScript*", several participants claimed. They did not seem to view themselves as technology experts, but instead as UX design experts who had great comfort working with a variety of technologies, and AI was one of these.

Instead, the participants understand AI mostly via designerly abstractions and examplars. We found almost nothing in common with the way designers spoke about AI and the way education materials meant to teach AI for engineers. Participants rarely spoke of AI in technical terms. For example, they never talked about supervised or unsupervised learning (common starting point for teaching AI). Instead, they appeared to think with, and work with, abstractions; simple insights about an AI capability had had implicitly linked with generating value for users. These were much more abstract than design patterns. More similar to design patterns, they often used exemplars to communicate these abstractions.

> Q: What does machine learning do?
> P7: *Some try to recognize intent, a bit like auto-correct. Some are intent prediction like Clippy. Anyone who is working on assistive technology, is working on some class of that problem.*

In the excerpt above, P7 describes AI using abstractions of its capabilities: recognize intents and predict intents. He then ground these capabilities through the use of exemplars: auto-correct and Clippy. We found this manner of describing AI across most participants.

Participants most often described the capabilities of AI as it related to the user's utility. Their abstractions narrowly oriented towards both the users and scenarios related to their designs. They never spoke of general taxonomies of AI functionality or specific algorithms. Some examples:

> *We use machine learning so that we can build something that can personalize for a lot of people.* (P3)
> *In consumer tech, we try to raise the level of abstraction [of user commands] rather than doing everything manually.* (P7)
> *Machine learning gets users directly into the task they really need to do.* (P1)
> *We are doing a recommendation system of sorts. As a product designer [not a technologist], I think about that as how can we show an evolving relationship between a user and our serviceâĂę [I want users] foreseeing our relationship improve, where the relationship is the recommendations we are giving them.* (P14)

The abstractions almost always appeared with exemplars. The abstractions served as a general insight about an AI capability and provided an understanding of how it worked. The design exemplars provided specific interaction possibilities and a glimpse of a possible felt experience. In our interviews, participants frequently referenced widely-known exemplars including Clippy, autocorrect, email spam filtering, and Tay the Twitter bot. They used these to help describe the capabilities of AI; the form, function, and user experience; and the potential breakdowns that might occur.

Participants actively grew their working set of abstractions and exemplars. They looked for these in both online articles and in competitors' products. One participant recalled that she carried around many conversational virtual assistants so that she could learn the feedback different products gave in response to the same instruction, in different contexts.

The number and variety of exemplars participants used varied wildly, and those with the largest working sets seemed to be the most successful and comfortable at using AI to enhance UX. Participants working at AI-focused organizations had a significant advantage in building their working set of exemplars and abstractions. These organizations had data scientists frequently giving demos as one way to sensitize their design teams to emerging AI capabilities. "*So many people demo for me. I don't even know whom to call if I have an idea and want to consult a data scientist.*" Participants at smaller or less AI-focused organizations had much more limited access to data scientists. They appeared to consider and propose fewer design alternatives. They were also more likely to use only the most familiar interaction forms, such as recommender or reminder, when describing their process.

3.2.2 How Experienced Designers Work With AI

Participants shared that working with AI took much longer than when designing other UX products and services. We wanted to synthesize the actual time span AI projects required. However, we found that none of our participants had worked on an AI product from its initial ideation until its final release. We could see that some AI projects had lasted for more than four years.

> *ML has a different time-frame for design iteration. Longer initial development, but then ongoing iteration. It felt like building a feature versus building a framework. When you ship it, it's not the end of it. They cycle, and data drives the next step.* (P3)

A consolidation of participants' design process narratives revealed an AI design process. It starts with a long preparatory stage (stage 1), during which designers and data scientists identify a design goal that is both technically viable and that appears to have a high likelihood of improving the user's experience. Once a team had settled on the design goal, scientists and developers developed the system and designers crafted its interaction design (stage 2). In stage 3, they together invested in frequent iterative releases and assessments in order to improve user adoption.

All participants described stage 2 in their AI projects. Stage 2 is a central stage that all projects that participants described went through. Stage 1 happened only at the large, AI-focused organizations. Few projects had made it to stage 3. Below I describe three distinct design stages.

Stage 1. New AI design seemed to only happen within the AI-focused organizations. It started with a long, preparatory stage involving two activities. First. participants collected log data form current services their organization provided. Participants examined the log with sensitivity to specific patterns. They often seemed to search for patterns they thought might be there more than they would data mine to discover unexpected patterns. They imagined what user behavior might be worth learning and what learned interactions might be valuable for users. One designer described this process as "*just me being really excited about the product*" (P14).

Second, participants would share these inchoate ideas with data scientists. This happened informally, outside of any specific development projects, and they had iterative interactions, where they might return to the same data scientists with new or refined ideas. They noted that the data scientists had a very different view of scenario-based design. They used scenarios to validate what an idea might be instead of using scenarios to generate and refine new ideas. "*Data scientists use scenarios to validate designers' assumptions about how the product should work, [and then] toss back ideas*".

The conversation between designers and data scientists focused on identifying a design goal worth pursuing. Often this would lead the discussion away from working with AI. The discussions focused on coming up with a "good enough" idea, and it did not address either its exact technical feasibility or experiential quality. Participants particularly pushed the data scientists to understand what might be technically possible.

> *(I) framed the questions not as do you know what would work, but in your gut, do you think this would be possible. Possible on a scale of 1-10.* (P8)
> *The level of detail I'll need to discuss with [the data scientists] is understanding the capability of what could be possible. I didn't need to get into specific detail about it.* (P3)
> *Those [design ideas] are a series of aspirations. Rather than saying what data do we have [...] Could we challenge the data team to figure out how to get close to that?* (P9)

The collaboration at this stage focused on co-evolving a shared vision between the two areas of expertise, an ideal user experience that was worth pursuing and AI could potentially help achieve. This shared vision often took form of a unique abstraction of AI capabilities that emerged out of both UX and AI. Instead of saying AI or ML, designers and

data scientists might select an expression better suited to the context of their product. For example, some used "affinity" to describe the match between a user and a piece of content. Others used "personalization" to describe the intended, evolution of the relationship between the product and the user. Participants characterized these discussions as a chance for both sides to learn from the others' expertise.

> *There is no such a framework or something, but I think later there is a kind of an acknowledgment when we talk about "personalization"... An acknowledgment that a more personalized experience is a better experience that one is less likely to walk away from.* (P14)
> *I gained a better understanding of the capabilities of the algorithms. The data scientist gained a better understanding of what was worth pursuing.* (P3)

Stage 2. Stage 1 produced preliminary understanding of the data, and it established a shared vision between the designers and the data scientists that met the company's goals. Stage 2 focused on refining this vision. In many organizations, this was the stage when UX designers were invited to join a project. During stage 2, participants described developing "*a funnel of visions, a funnel of what exists and what is possible in the company*" (P8). This would advance towards defining a single, valuable AI feature that was both experientially valuable and technically feasible.

> *It's definitely an ongoing process. We had a lot of basic things to get working. [...] Once we get that improved, then we will probably be adding more ML stack onto the system.* (P14)
> *Design is always about a new product and a complete, larger vision. But that doesn't make it easy to build. We chunk. We talk about stages of development.* (P8)

Once a product idea was clearly formulated and agreed upon, participants shared that they would next put the required resources in place. Participants would start to design the interactions in parallel with the technical development. Participants explored many possibilities. They also spoke about assessing their interaction designs based on following criteria:

- *Could users produce effective labels needed to train the AI system?*
- *Could users make sense of the AI inference [or adaptation] and did they view it as valuable?*
- *Could users easily recover from AI [inference] errors?*

Participants used a combination of traditional UX methods such as user studies, sketching, and usability testing. They also engaged in continued negotiations with technologists. Collectively, these activities allowed them to craft the user experience. This iterative process continued until the interaction design had matured to a point it could be handed over to a front-end development team.

Today, most new features for online products and services only see broad release after passing a series of A/B tests, and the same process was used for AI features. These tests are meant to reduce the risk of user abandonment and/or reduction in new user conversion rates. Almost all participants spoke of A/B testing as a critical part of improving their design, and this related to both improving the algorithms' performance and improving the interaction. In most cases, the participants shared that they offered users the option to turn a new AI feature off. This appeared to help them view passing of the A/B tests as less challenging.

Stage 3. Few of the projects described by participants made it to stage 3. One notable exception was P7's worked on a message classifier. At the beginning of stage 3, they worked to improve the poor adoption rate. This became "*a really key moment in thinking about design of systems like this*". Stage 3 had to do with refinement that could make a project successful.

> *We had a classifier that predicts whether or not an incoming message is important âĂę and the performance is scary good. âĂę However, a vast majority of users didn't turn the feature on or soon opted out. When you dug into why that was, people would say "I don't even know what's important to me, how am I gonna trust the ML system to know it's important to me?"*
> *The core insight my research team had was that people are trying to figure out whether they are gonna trust that system, and the way they figure it out to think of it as a person. If I hand a stack of messages to a person, someone I don't know, is it possible that person could figure out with some accuracy if that message is important to me. Most people's take was "no, I barely know it myself".*

Stage 3 sent the UX designers into the field to investigate how people used their system in order to improve the design. The team improved the design by moving away from a binary classification (important or not important) to a small set of categories. The new design made it clear why something might not be important by classifying it as "promotional materials," something the team felt any human would be able to do. The new design used the same AI technology and proved to be a huge success.

The few participants who had projects reach stage 3 stated that such a redesign was inevitable because some mental model issues are difficult to catch before products launch. They felt they "*have to do this level of work on the design side*" (P2,7,13) so that users cannot only recover from the errors, but can understand the errors, and at the same time preserve some level of trust in the system.

> P1 describing why design problems for a chatbot auto-reply feature were unforeseeable: *We help you [users] say that you already gonna say. We do it a bit faster. We are actually influencing what you are saying, but not predicting what you are saying.*

3.2.3 Emergent Design Practice in Data-Driven Culture

We probed participants on what they needed to be effective and what UX students should know in order to effectively envision AI innovations. Their responses collectively revealed an acknowledgement that they all worked in data centric environments, and that designers needed to embrace this data centricity in order to have impact. They spoke of the importance of learning to speak the language of quantitative data and data science (e.g., telemetry, analytics, A/B testing, covariance, correlation). In the pre-interview surveys, participants responded that they worked with these concepts constantly. "*This is how engineers measure and businesses do things. This can influence how you design.*"

To help embrace a more data-centric culture, participants shared that they engaged in new design activities. They used a combination of qualitative and quantitative methods to advocate for their design. Specifically, they developed new skills around collecting telemetry data (data remotely collected from current products and services), and they generated data visualizations as one way of making sense of their data. In addition to working with data, participants shared that their collaborating data scientists share their role as a user advocate and as facilitator of incorporating domain experts' insights; something that did not happen with other, technical development efforts.

Designing Telemetry Data Participants frequently asked AI specialists about how people use the product, how they feel about it, and what problems they had that could be rectified by design. To help answer these questions, participants shared that they designed telemetry systems so their products can better track users' interactions. They used the data to capture meaningful and accurate snapshots of user behaviors.

Designers began to define their design goals in relation to the user behavioral data collected telemetrically, in the form of a "*matrix for a good experience*" (P2,8,12). This matrix would later be used to measure the success of their designs

through a suite of A/B tests. Designers use these matrices to "*influence the engineers to think a different way, not in terms of single A/B test, but a suite of tests.*" The telemetry data thus became a space of exchange between designers and data scientists. The data helped to expand the narrow scope of A/B tests and to more holistically address user experience over a larger course of interactions.

Designing Data Visualizations Participants worked to interpret user behavior data. Most utilized customized dashboards that translated mundane log data snapshots into user stories and insights. Several taught themselves to create data visualization tools, creating visualizations that used by data scientists, engineers, and fellow designers in their organizations. They designed dashboards and visualizations to combine an immersive and an analytical way of understanding, so that the quantitative analysis of user behavior "*do not privilege data scientists*".

> *Data scientists have [...] kernels and clusters, and we are good at telling rich, compelling user stories. How can we look at hundreds and thousands of attempts, and also reach out to inquire more about what happened, what the breakdown was?* (P14)

A few designers further utilized these data visualizations as vehicles for conducting user studies. One participant recalled that the first thing he did when joining the project was to list "*a series of questions about user behavior that data analytics can answer*" (P8). They then bookmarked the corresponding data matrix on the telemetry analytic tool the team used. Every morning they checked the new incoming log data against this matrix. When they noticed an intriguing user behavior, they emailed the user to inquire the back story, details, and sometimes set up follow-up interviews. They labeled this method as "*qualitative study in a quantitative scale.*"

Advocating for Users Along with Domain Experts Besides data scientists, designers also worked closely with domain experts during all phases of a AI project, especially projects that design special-purpose applications. For example, P12 designed a reminder feature of a social network product in collaboration with psychologists who specialized in human perception and memory; P15 designed an intelligent tutoring system on biology in collaboration with middle school teachers and biologists.

While cross-disciplinary collaboration is not new nor unique to AI projects, participants suggested that when working with AI, data scientists and domain experts overlapped with the responsibility in their work to make sense of and predict user behavior. They were no longer the sole user advocates. Most participants repeatedly stressed the unique skills designers brought to the table in relation to other human-centered ML roles; some designers also adapted new methods to manage and communicate their designs:

> *I embedded an information-motivation-behavior framework (in user data analysis). It showed everyone that just (user) information is not enough.* (P8)
> *Designers have domain knowledge about how people typically respond to certain things, how easy is it to change a certain behavior. Designers helping with shaping the experience, with the missing part, the things you have not thought about.* (P9)

3.2.4 Summary

The main goal of this thesis is to enable designers to leverage AI in their designs, and to conceive of entirely new ways to use AI. The practice of experienced designers provides a point of reference toward this goal. For example, HCI and UX educators can better prepare students to more successfully enter industry by training them to work and think with data and providing opportunities for them to collaborate with data scientists. The findings also highlighted a set of pragmatic questions designers grappled with in practice, for example:

- How may designers conceptualize AI's capabilities?
- How may designers sketch AI-mediated interactions?
- How may designers effectively communicate the design ideas underlying AI interactions to users, other designers and collaborating AI engineers?
- Do designers need technical AI knowledge to design? If so, in what aspect and to what extent?

3.3 Discussion and Reflection

In the two exploratory studies, I searched for insights from both research literature and experienced practitioners to better understand AI's design space and design challenges. Findings of these two studies offered a set of tools and methods that other designers can directly re-use, such as the AI research topic landscape, the inference-acting schema, the domain-specific abstractions of its capabilities, and UX value channels. We see opportunities around disseminating these research outcomes and best practices in industry to help UX designers better work with AI.

These studies also draw attention to a number of related issues on which these emergent knowledge and methods of designing have started to destabilize the current UX practice. Underlie these issues are the differences between AI and other common technology materials, and therefore merit further study.

How should designers develop abstractions of AI's technical capabilities and examplars?

In order to get a grip on AI's design potentials, we proposed a set of abstractions to scaffold its technical capabilities, interactions, and experiential values. The designers we interviewed also comprehend AI in notably different ways than its textbook definitions; they understand AI largely through abstractions and examplars. Particularly, the abstractions served as a frame through which designers reflected on the design challenges at hand and made new assertions about how AI might provide value for users.

These abstractions not only freed designers from grappling with technical limitations when sketching, empowering them to envision AI applications that moved beyond stereotypical forms. They served as boundary objects, allowing designers to discuss what users value with data scientists and to address issues of context. They also fostered new design ideas, serving as bridges between technical capabilities and design possibilities. HCI researchers could potentially help designers grasp could also grasp the design possibilities by creating a ready repository of AI abstractions. We also see opportunities in advancing the abstractions by creating many instantiations of the same abstraction as a way of investigating the opportunity space.

Do Designers Need Technical AI Knowledge?

The interview study showed that designers were *capable* of engaging in a reflective conversation with AI without a formal AI education. Yet it remains an open question whether a more advanced understanding of AI would inspire new design perspectives and new designs. On the one hand, design research has long promoted a "designerly way of knowing and doing" âĂŤ the idea that designers can work with technologies and situations that they do not fully understand [10,84]. More investigation is needed to better define what is a "good enough" understanding of AI, and to understand when a more detailed knowledge of AI might help.

Should the user-centered design process change? Previous work shows some designers do not know how to bring their UX expertise to bear on AI [30]. The above studies offered two distinct possible solutions; One is match-making, the other is a divide-and-conquer approach. Unfortunately, match-making as a design method is under-investigated and under-utilized. Its validity and generalizability are difficult to assess.

The designer practitioners in the interview study followed a different design process. They first identified the right problem for AI to solve, validated its technical feasibility, and then iterated on the diverging-converging ideation process

to craft the right manifestation of AI functionality. We found this overall approach to be similar to descriptions of the lean startup with a focus on making a minimal viable product [79]. They were similar in the way teams would assess the viability of an idea and pivot when it seemed to fail.

More research is needed to reflect on and improve the processes we explored and observed. At a higher level, the procedural knowledge of designing ML marks a clear space for HCI/design research. Existing work has offered valuable declarative knowledge and conceptual understandings of ML from a design perspective. Embedding this growing body of new knowledge into organizational and procedural contexts opens up new research opportunities and promises real impact on UX practice.

How should UX and AI technical expertise collaborate? Echoing findings from previous HCI research [41], our interviews captured an intimate, constant, cross-disciplinary collaboration when creating AI products. This is somewhat different from the design and technical collaborations found in a traditional UX design [26], where designers typically deliver a fully formed design to a technical team to implement. We see opportunities for new collaboration tools that help designers better work with data scientists, and tools that support designers who lack constant access to capable data scientists.

How can HCI researchers synthesize emergent solutions to AI's design challenges? Let us step back for a moment and more broadly consider ways to advancing on the above research questions. The two studies presented in this chapter share one common limitation, that their findings – various ways of understanding AI, design methods, tools and processes – lacked evaluation. The effectiveness of such emergent design knowledge in inspire AI's design innovations is a very difficult thing to measure because it is nearly impossible to control for. Consequently, it is very difficult separate the findings that are specific to the case from those generalizable to other instances of AI.

Design knowledge arises from, and in response to, concrete problems and situations [44,83]. Yet without a framework of the problem space, HCI and design researchers cannot easily synthesize the emergent knowledge or inform new designs beyond the particular design instances they have produced. This realization forced me to first outline what are we talking about when talking about working with AI as a design material, before moving on to investigating the research questions raised above.

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4 PROFILING AI AS A DESIGN MATERIAL: AN INITIAL FRAMEWORK

In this chapter, I propose a useful framework for conceptualizing AI as a design material. Despite there being an rapidly growing interest in UX of AI, there have been surprisingly few serious attempts at systematically accounting AI's materiality, or what it means for UX design. Hereby I take prior literature on technology as design material as backdrop, and structure the framework around two interrelated yet distinctive characteristics of AI: One is its multi-layered capabilities, that AI can possess distinctive capabilities in the lab, in the field, and when being encountered by individual users; The other is the decoupling of capabilities and system behaviors. Together they form a useful framework toward unpacking the design opportunities and challenges AI poses.

4.1 Theoretical Backdrop: Technology as UX Design Material

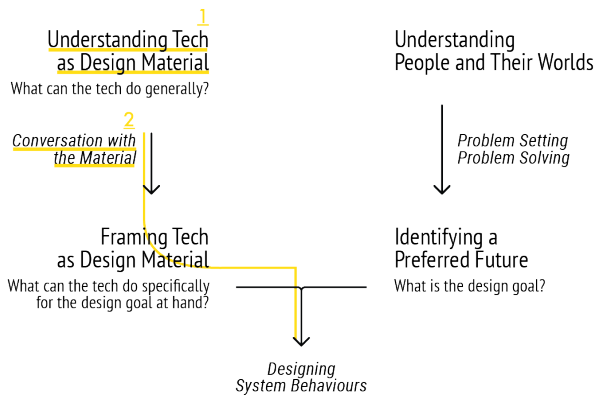

Fig. 7. How design material fits into design processes. The left branch refers the design activities for getting to know the technology materials, while the right refers those for understanding the world in which the technology will operate in. Prior research [99] also has drawn particular attention to the distinction between designers' felt understanding of a design material (1) and their situated conversations with the material (2).

Fallman describes the role of design in HCI as the act of unfolding a previously nonexistent artifact from the various bits and pieces gathered in the process of research. Fieldwork, theory, and evaluation results all provide input to this process, but do not by themselves bring the new, coherent whole into being. "*For the latter, there is only design.*" [33] Schön characterizes this process of unfolding a coherent whole as designers' "reflective conversation" with materials of the design situation [85].

In addition, Schön highlights that the materials of the design situation are designers' subject choices. Through active appreciation of the design situation, designers construct and reconstruct the materials with which they deal, thereby creating a design world within which they function [85].

Research further makes distinctions among engineering, designing technology at an application level and at a material level. Engineers create new technology that allows new capabilities, while designers create new assemblies of known capabilities [64]. When designing at an application level, designers give novel forms and expressions to the technology's given functionality; When taking technology as a design material, designers work to understand its capabilities and limits, actively explore what new aesthetics, functionality or experiences it may afford [76,99].

Understanding a technology as design material is not just a matter of properly describing its capabilities, but also of engaging with them in respect to designers' certain interest or perspective under specific design situations [95]. Sketching and prototyping exemplify two archetypal processes through which this process can unfold, that designers reconfigure, amplify, or de-contextualize various factors of the technology to see what new design possibilities may emerge [85]. Again, take cardboard as an example. Designers may have some ideas of its properties such as weight and strength, yet it is through bending, folding, tearing it and observing how it responds that the designers develop a felt understanding of cardboard. They carry out reflective conversations with the technology and in the process develop a tacit understanding of what it might become.

Figure 7 outlines these theoretical constructs, providing a backdrop for understanding AI as a material for UX design.

4.2 Key Characteristics of AI

Taking a step toward unpacking how AI differs from other technology materials, I articulate two key factors that shape how AI is designed and experienced: 1) AI's multi-layered capabilities and 2) the decoupling of its capabilities and system behaviors.

4.2.1 Distinct Capabilities in the Lab, in the Field, and as Experienced by Users

The first concerns AI's technical capabilities and limits, that is, what an AI system can infer, how good the inference is likely to be, and what errors it is likely to make. AI's capabilities lie in the interplay between algorithms and data. Algorithms hold the capabilities of making inferences; Data manifest this capability.

Fig. 8. AI's three-layered capabilities. The distinctive capabilities in lab before deployment, in the field, and as different users encounter it make AI a particularly powerful yet particularly challenging design material.

Because of data, AI's capabilities take shape in three stages; the capabilities evolve as new data come in (Figure 8). Before an intelligent product or service is initially deployed, AI algorithms learn from a self-contained body of data ("lab data"); After the system gets deployed and users start using the system, the traces of user interactions ("field data") become an additional set of data that feed the algorithms and power the continued evolution of their capabilities. This evolution is almost always uneven across different users and use contexts, since the interaction traces are rarely evenly distributed; Within this range of capabilities that AI holds, a user's "local data" (such as use history, time and location, etc.) determines the exact performance of the system that a particular user will encounter at a particular time.

As a result, the same AI system can exhibit distinctive capabilities at each stage. Consider a voice recognition system that can accurately translate 90% of some given audio files in the lab. In the open world, the accuracy may become 99% accurate with some accents and 30% with others, depending on the accents of its users. The exact accuracy that a particular user will encounter at a particular time (the "localized capability" in Figure 8) may be anywhere in-between 30-99%, conditioned by the accent the user speaks in, ambient noise and numerous other factors.

The multilayered capabilities make AI less like most common design materials. For example, cardboard becomes discolored and frail when containing 10% water content. This capability and limitation is consistent in the lab, in the real world, and for any person interacting with it. Even though a cardboard-made *product* may exhibit different capabilities, for example with waterproof paint, the capability of the *material* does not change. Same as most other technological design materials. Consider a chatbot that replies "A" whenever an user says "B", as an example of manually-specified-rule-based, non-AI software. Its capability is also consistent in the lab, in the real world, and for any users.

4.2.2 The Decoupling of Capabilities and System behaviors

Let us revisit the distinctions among engineering, designing technology at an application level and as a design material. Engineers create *new* technical capabilities; Designers give form to a *given* capability when designing at an application

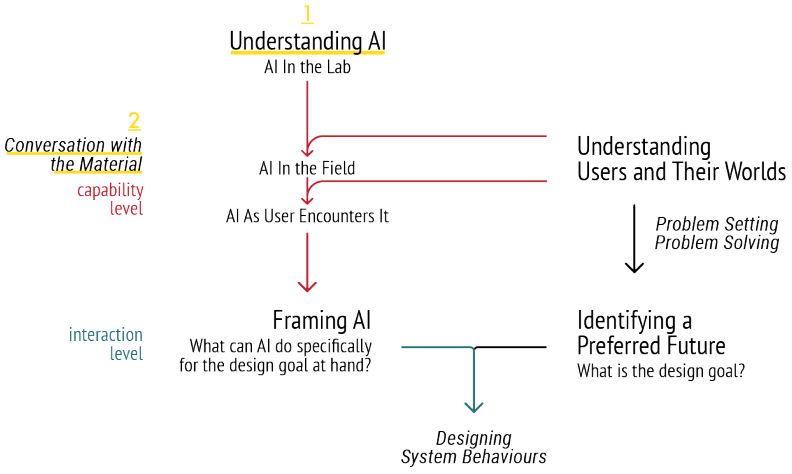

Fig. 9. An initial framework for conceptualizing AI as a UX design material.

level; When taking the technology as a material, designers have a reflective conversation with its various capabilities and actively explore new interaction and experiential possibilities that are meaningful to the design problem at hand.

The boundaries among these notions are blurred in the context of AI. AI capabilities evolve and diverge autonomously, making it difficult to distinguish the new capabilities from existing ones, or to distinguish one given capability from a contingent one. Hereby it becomes necessary to restate the difference between technical advances and design innovations in the context of AI, and to more explicitly decouple of the notions of designing AI capabilities and system behaviors.

- Engineering efforts improve or create new AI capabilities, such that AI systems make better inferences on in-lab data and/or on unseen data in the field. AI engineers work to understand user behaviors and contexts in order to better understand data and improve algorithm capabilities, while designers do so in order to better situate existing AI capabilities into these contexts.
- When taking AI as a design material (Figure 9 red lines), designers actively explore what AI can or cannot do for the purpose of framing and reframing how AI will contribute to UX.
- When designing AI at an interaction level (green lines), designers take the framing of "how AI will contribute to UX" as a given and orchestrate concrete system behaviors and interactions. Noteworthily, designing at this level may also require designers to engage with AI capabilities at all layers. Designers do so to better understand AI system behaviors, rather than for the purpose of reframing the value AI will provide for UX.

4.3 Using the Framework

So far I have identified AI's multilayered capabilities and, consequently, the layered conversations with AI capabilities and interactions as two thematic qualities affecting how AI is designed and experienced. Integrating them into the aforementioned overall process of designing technology as material creates a useful framework for conceptualizing AI as a design material (Figure 9). Specifically, this framework is useful in three respects:

- *Analytical leverage*: It helps scaffold and synthesize a full range of opportunities and challenges of designing AI;
- *Explanatory power*: It helps specify the contribution and limits of emergent design knowledge;

- *Constructive potential*: It provides more value than objectively organizing known design challenges. It offers valuable new insights into ways to addressing the challenges.

4.3.1 Scaffolding AI's Design Opportunities and Challenges

The multi-layered capabilities make it explicit why AI is a particularly rich design material. Even at an in-lab level, for example, AI can draw complex and unintelligible insights from human utterances and comprehend natural language, enabling otherwise impossible forms of interactions. Moving out of labs to the open world, AI can further manifest dynamically evolving and massively divergent system interactions. The design opportunities in utilizing user and open-world data have already spurred a round of AI design innovation (Figure 9 red lines). Design opportunities further lie in translating AI's open-world capabilities into situated user encounters. For example, the Local Data Design Lab at Georgia Institute of Technology invited city residents to interpret the data of their city, exploring design opportunities of highly localized experiences using big data [6].

In this light, the framework provides a concrete way for HCI designers and researchers to conceptualize the full design space of AI in their own research and design. Against the backdrop of this framework, future HCI research in various subject domains can also take stock of what is known and to identify under-explored design space as a basis for future design and research endeavor.

In addition, the framework offers a set of themes around which we can summarize a full range of challenges of designing AI. For example, immediately visible from Figure 9 is that designing AI requires more frequent exchanges between designers' conversation with the technology and their understanding of the real world. Designers cannot orchestrate interactions of AI without a grip on its capabilities at the start, yet understanding AI capabilities in turn requires an understanding of the landscape and nuances of user behavior and contexts. These dependencies challenge the typical workflow of user-centered design [26], where designers work to understand users in the field, play with the technology in workshops or labs, and deliver a fully formed design to a technical team to implement. This also explains the new design practices we observed in the industry, such as a constant and tight collaboration with UX and AI technical expertise, and a design-and-development process that resembles lean startup.

The framework therefore offers a valuable conceptual tool for researchers to untangle the many interrelated challenges of designing AI. Design researchers can prioritize their research efforts accordingly and advance on the challenges of their choice by purposefully putting themselves in selected design situations and confronting these challenges. They can more effectively build upwards toward its overall solutions, by solving a set of smaller problems they encountered in distinctive design situations.

4.3.2 Understanding AI's Emergent Design Knowledge

HCI and design researchers have so far created many AI-infused systems and experiences, and in the process, identified many valuable insights and emergent design methods and tools. However, they cannot easily separate their findings that are specific to the system/algorithm/use contexts they have focused on from those generalizable to other instances of AI. They cannot easily articulate the contribution of their work to the broader challenges of designing AI.

By providing structure to the challenges of designing AI, the framework offers researchers intuitions about the contributions and limits of their work. Consider the various methods of "prototyping AI" that HCI research has recently proposed. Interactive ML tools such as Wekinator [2] support rapid prototyping by allowing designers to experiment the embedded algorithms' capabilities with varying kinds of input data; Designers can therefore develop a felt understanding of the pre-identified *lab capability* as experienced by different users and in various use contexts. These tools do not take

into account how these various data may shift these capabilities collectively and over time. WoZ platforms such as the Delft toolkit [96] and Suede [57] also promises rapid prototyping of AI. AI systems simulated by WoZ can integrate the collective behaviors of users (field capability) and at the same time interact with them (AI capability as user encounters it). However, it is difficult to fully capture AI's lab capabilities and limits with WoZ.

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5 THE FRAMEWORK AS A RESEARCH THROUGH DESIGN TESTBED

This chapter describes two sets of RtD projects. One is in the context of clinical ML; The other in designing natural language intelligence (NLP) systems. I provide first-person accounts of these design processes, and more importantly, my "reflective conversations" with AI. I share the emergent solutions and failed attempts to the aforementioned challenges of designing AI; Draw on these experiences, I also identify previously unknown challenges and more broadly reflect on the framework of AI as a design material.

5.1 Designing a Clinical Decision Support Tool for Artificial Heart Implant

5.1.1 Background

The idea of leveraging machine intelligence in healthcare in the form of *decision support tools* (DSTs) has fascinated healthcare and AI researchers for decades. These tools promise improved healthcare quality through complementary insights on patient diagnosis, treatment options, and likely prognosis. In recent years, the adoption of electronic medical records along with advances in big data technologies has created the perfect environment for algorithm-powered DSTs to impact clinical practice.

Despite success in labs, the vast majority of DSTs failed when they moved to clinical practice [32,53,54]. Clinicians rarely use them [29,32,100]. In a review of clinically deployed DSTs, healthcare researchers have speculated that the lack of HCI consideration in the design of these systems might be the main cause of these failures rather than poor technical performance [87,92]. This includes a lack of consideration for clinicians' workflow and the role AI currently plays in clinical practice [68,98]. However, little to no work in the field of HCI has investigated these issues or proposed how intelligent DSTs should be integrated into care environments.

I have been collaborating with biomedical engineering researchers on the design of a DST supporting the decision to implant an artificial heart. The artificial heart, VAD (ventricular assist device), is an implantable electro-mechanical device used to partially replace heart function. For many end-stage heart failure patients who are not eligible for or able to receive a heart transplant, VADs offer the only chance to extend their lives. Unfortunately, many patients who received VADs die shortly after the implant [12]. In this light, a DST that can predict the likely trajectory a patient will take post-implant, should help identify the patients who are mostly likely to benefit from the therapy.

The DST for VAD implant decision will be a prognostic DST using data mining to help clinicians make a good destination therapy choice. Like almost all other prognostic DSTs, it currently takes a context-less, prototypical form: It takes in a list of patient condition measures and produces an individualized prediction of patient trajectory, including likely survival and other post-surgical risks [11].

Given the previous failures of DST deployment and the wide gap between DST technology and clinical reality, we had three goals with this project.

(1) We wanted to better understand the clinical decision process around a VAD implant and the clinical reality in general, so as to identify the key touch points where we might situate a prognostic DST that clinicians would find useful in their practice.
(2) We wanted to offer concrete solution(s) to the long-standing challenge of effectively situating DSTs in clinical practice, and probe clinicians' responses to a DST situated in their workflow.
(3) We wanted to explore whether the insights and the design would likely to generalize to other clinical decisions beyond VADs.

5.1.2 Field Study: Understanding Clinicians and the Clinical Reality

We first conducted a first study [107]. We wanted to understand the clinical decision process around a VAD implant, including the participants, their work practices, the contexts where the decisions get made, and other critical factors that influence the decision. We wanted to identify the key touch points where we might situate a prognostic DST that clinicians would find useful in their practice.

We interviewed and observed clinicians caring for VAD patients at three different implant centers all in the United States. The three facilities vary geographically and in scale, and their performance rankings range from top 5 to top 60 in the United States. We conducted IRB approved interviews with a total of 24 VAD clinical team members from 3 hospitals, covering many different roles and statuses that participate in decision-making. We then analyzed our data using affinity diagrams and a service blueprint of the decision paths different patients follow.

The field study identified many barriers that could negatively impact the use and perceived value of a prognostic DTS situated in VAD implant hospitals:

- *Attitudinal barrier:* First and most importantly, clinicians we interacted with have no desire to use a DST. They view the decision to implant a VAD as easy. As long as patients have no definitive exclusion conditions, they will all get a VAD after failing on an identical, escalating sequence of less aggressive treatments.
- *Need Barrier* Clinicians' experience with current tools like the VAD risk models has in no way provided any confidence that DST or other intelligent systems can provide valuable new data. It is unlikely that they would explicitly use any DST until they perceive a need and until they trust these systems can deliver value. A better DST would have the explicit goal of helping clinicians feel they are doing better work, and not necessarily automating the part of work that makes them feel like an expert.
- *Social Barrier* The lack of a real consideration for social context in the design of DSTs can be a significant deficit. The hierarchical but collaborative clinical culture poses a two-fold challenge for DST use. First, decision makers (physicians) and computer users (the midlevel) rarely overlap at the point of decision-making. Second, physicians have great trust in their social network of other physicians, who help them make more difficult decisions. It seems unlikely they will move towards computational support and away from social support when things are difficult.
- *Informational Barrier* We observed a mismatch between clinicians' information needs and DST's information flow. The commonly assumed function of a prognostic DST is to predict the likely trajectory a patient will take based on a list of quantitative measures. Our patient trajectory paths demonstrate that none of the major decision breakdowns happens in this prediction.

At the input end, DSTs take in quantitative and explicit inputs, while challenging decisions are often characterized by unavailable or ambiguous medical and/or social evidence. Clinicians are unlikely to use a tool that only does the easiest part of their job; telling them a textbook case is, "textbook". Even if they approach the system when facing a difficult case, they might find it difficult to fill in some of the blanks, such as diagnosis for an emergency-room-path patient. They might find the information that most concerns them is not captured in the prediction, such as the patient's home life and social support, which are critical and difficult factors most often not captured in the medical history.

In terms of DST output, physicians need support for action taking. Consultation between cardiologist and surgeon best captures this: *Is this case too risky to operate on? No? Ok, then do it.* A probabilistic prediction can be obscure in telling whether to execute a therapy or not, to do it now or to "wait and see". DSTs only predict outcomes of "conducting a therapy now", with little sense of waiting and seeing.

These observations forced us to reflect on the traditional forms most prognostic DSTs take. Most require clinicians to recognize when computational advice would be useful and then make an explicit effort to access a DST [68]. In addition, most imagine a single decision maker participating in making the decision at a single time and place [93].

From our field observations, We draw three implications to inform and inspire DST designers in addressing adoption barriers as well as in exploring new design possibilities: embrace context, make the decision process a design material, and blend human and machine intelligence.

- **Embracing the Richness of Clinical Context.** A core goal of our study was to explore how a DST might better fit in clinical workflow and social context. Our study revealed rich details illustrating the context, which open up new opportunities for DST designs to integrate and leverage.

  DSTs should be integrated into EMRs (electronic medical records) or they should at least automatically take in EMR data as inputs. They need to minimize input of data due to clinicians' frequent hand washing and lack of time spent in front of a computer.

  The fact that seasoned physicians do not perceive a need for decision support suggests that DSTs have to make an effort before they can reach and convince these decision-makers. Designers should leverage the midlevels who more frequently use EMRs as a channel for delivery of decision support. Designers should approach this with some caution, as this may disrupt the hierarchical decision structure that is in place, but it could also positively elevate the role midlevels play in decision-making, thus encouraging them to participate. Central to this point is that establishing credibility and value across all members of the implant decision team should be a primary design goal.

  In addition to midlevels, we also view the weekly implant meetings as an opportunity. A DST that wants to demonstrate value might automate the process of preparing patient information for this meeting. By automating the tedious information retrieval tasks, a DST could ease its recommendations into the discussion materials that the whole implant team reviews.
- **Decision Process As Design Material** Our finding illustrates that a healthcare trajectory, as well as its decision process, is pushed forward by a string of treatment escalations. We believe this new perspective on decision-making inspires a new theme in the DST design space: Decision support along the trajectory.

  A VAD implant decision is anchored by a set of many smaller decisions that clarify and optimize patient conditions. Our illustrations of patients' journeys indicate that a breakdown at any of these steps can negatively limit therapeutic choice. We see a real opportunity for DSTs to provide much more integrated support. Currently, DSTs take only one form. They support making right *or* good decisions; they either help making a diagnosis, or a treatment choice, or

a prognosis. Our findings prompt DST designers to consider combining a range of DST components with various forms and functions in order to support many small decisions that often lead up to a major clinic decision.

We observed that initiating timely care escalation has a crucial and direct impact on VAD decision quality. The late referral path offers a perfect example. DSTs might be able to improve patient outcomes by supporting physicians across healthcare facilities with smart adjustment of patient clinic visit intervals, timely consideration of hospitalization, referral, and a formal workup for VAD. All along this process an intelligent system could be monitoring to make sure a patient does not arrive at an implant facility after it is already too late for an implant. We see these functions as particularly valuable for local or primary care doctors who do not specialize in VAD, but care for the majority of heart failure patients. In a broader sense, DSTs could help surface newer care options to primary care and local doctors who are not current on advances in sub-specialties.

A better clinical DST could prompt both patients and care providers to resolve fixable implant exclusions. DSTs could flag behavioral factors such as smoking, drinking, and a patient's BMI. In addition, it could also prompt upstream social workers, as well as patients and their families to address a lack of effective social support needed for post VAD life and a lack of insurance that would cover this expensive procedure. Solving these issues earlier in the decision process reduces the likelihood a patient might miss an implant window due to an exclusion criteria that could have been resolved.

Developing DSTs to support the management of a panoramic healthcare decision process marks a clear space for future research in both data science and HCI. We imagine prior research on care planning could be leveraged in support of this. Data scientists can explore healthcare process data and predicting longer-term treatment outcomes. We also suggest HCI researchers investigate the decision-making activities in local or primary care settings and further examine this concept.

- **Blending Human and Machine Intelligence.** Our findings highlight the attitudinal/informational barriers DSTs face. Currently clinicians have little motivation to use DSTs and many barriers stopping them. In our study, implant physicians expressed no need or desire to use any DST or risk model because they find no difficulty in making a VAD candidacy judgment. They also found DST data support for VAD decision inferior to fellow colleagues' input. This provokes us to critically re-consider the role of DST in decision-making tasks.

  In our study, VAD physicians reported that they know how to make a VAD decision. As trivial as it sounds, it is a missing perspective in DST literature that has instead focused on the clinicians as a source of errors, biases, overconfidence, and communication breakdowns. This assumption behind DST development and design, though not immediately evident in interfaces, perhaps seeds this attitudinal barrier. Many of our participants implied that makers of current prognostic systems want to replace their expertise with inhuman technology. Taking a lesson from early HCI work in participatory design, we need to make technical advances that skill workers instead of de-skilling them [9].

  Clearly there are opportunities for designing new interactions between DSTs and clinicians that work to integrate the abilities of both agents. One straightforward solution is to focus on more pliable forms of interactions, such as alerts and reminders. To date, one of the only successfully adopted DSTs has been alert systems. These systems require minimal user effort to manage and pose little disturbance on those who can make a correct decision.

  At a deeper level, the attitude and informational barriers have resulted from a simple fact that clinicians will not use DSTs for tasks they feel they can do better than a machine. There is a real need for new DST design that better allocates human and machine intelligence into different components of healthcare decisions. Clinicians might make better judgments than algorithms in synchronization of clinical evidence and social evaluations. Some patient situations

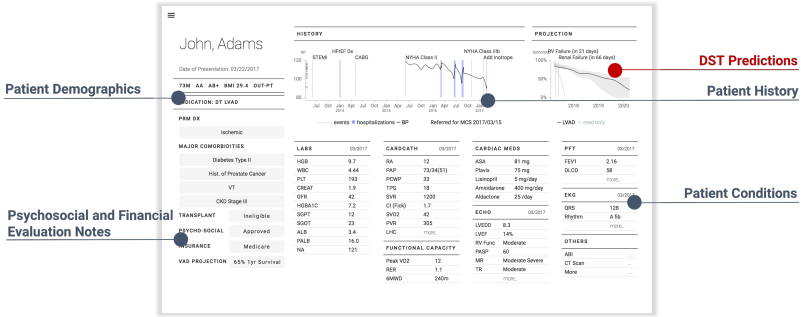

Fig. 10. The decision meeting slide design. We designed a DST that automatically generates decision-meeting slides for clinicians with subtly embedded machine prognostics at the top right corner.

reported by our participants suggest these difficult situations are where data-centered systems are less likely to offer helpful advice.

In other decision tasks, such as clarifying and monitoring patient condition as well as managing care escalation, machine intelligence can and should help. We see emerging opportunities in these spaces for DSTs to add value. For example, when facing an emergency-room-path patient with sparse data available, clinicians seem the most likely to benefit from the collective intelligence that is collated across many implant centers. Such cases might be the best opportunities for DSTs to gain trust from clinicians by addressing an actual situation where a need for support might be present. Clinicians might also value computational support in referral management. We observe inter-site clinician collaborations are not remotely as frequent and ripe as the ones between colleagues. Information technologies could potentially perform matching inter-site consultations much more precisely and catalyzing new referral relations much faster than the currently manual methods.

We also see opportunities for new DST designs that blend different clinicians and decision support components in decision space. Central to this implication is to make clinicians feel they are becoming better at their job, and to enhance clinical decision quality by leveraging advantages of both human and machine intelligence.

5.1.3 Design: Making Clinical AI Unremarkable

Embracing the rich context of the implant decision, we designed a radically new DST that automatically generates slides for the required decision meeting. The design embeds prognostic decision supports into the corner of decision meeting slides. We wanted decision makers to encounter the computational advice at a relevant time and place across the decision process, and we wanted this support to only slow them down for the few cases where the DST adds value to the decision.

This design draws inspiration from Tolmie et al.'s notion of Unremarkable Computing. They introduced the notion of unremarkable computing when discussing how ubiquitous computing should arrive and create its place in people's homes. They argued that technology can augment people's actions in ways that have a wealth of significance but seem unremarkable, because its interactions are "so highly situated, so fitting, so natural". They argued that home technology should not only be more intelligent, it should also be more subservient to people's daily routines. In doing so, the technology becomes part of the routines, part of the very glue of their everyday life.

We draw connections between this ambition and our aforementioned design goals. We also draw connections between this notion of routine and VAD decision making. While these are daunting life-and-death decisions, the implant decisions are part of a work routine for clinicians. To fit a DST into their practice, we need to make it subservient to the day-to-day decision-making workflow they engage in.

These orientations are very different from the convention of DST design in which decision supports are always available, waiting for clinicians to walk up and use at any point across the decision-making process. Instead, we wanted to tailor the DST for particular moments in the process, such that clinicians do not have to take pause and invent sequences of action anew. We wanted the DST to naturally augment the actions of decision making, rather than pulling the user away from doing their routine work.

5.1.4 Experience Prototyping and Field Evaluation

We shared the slide design at decision meetings, and we interviewed physicians and mid-level healthcare workers who play a critical role in VAD implant meetings to get more feedback. Finally, we spoke with physicians working on clinical decisions outside of VAD implant, probing whether this slide for decision meeting form might generalize to other critical, clinical decisions.

Designing DST to Augment Clinical Routine. Our findings suggest that the decision meeting offers an effective moment for clinicians to encounter decision support. Most clinicians involved in the VAD implant decision would likely encounter the DST output if it was included as part of an individual patient's information presented at the decision meeting. More importantly, similar opportunities to the decision meetings may lie in other time and place in existing clinical decision-making routine that is socially-aggregated, deliberative and shared across hospitals. Future research shall advance this work by systematically searching for such opportunities.

Intricacies of Designing a Right Level of Remarkableness. Both seasoned physicians and mid-levels expressed appreciation for DSTs that could slow them down "only when necessary". They liked this aspect of our design. However, we could not easily conclude whether our specific design had achieved this goal.

Clinicians shared that they could not draw on their experience of making critical clinical decisions seeing only patient data on paper. This presented the biggest barrier to assessing how clinicians might respond to a conflicting DST prediction. Patient history data alone did not give clinicians enough confidence to make an implant decision. Physicians described the meeting data as merely a surrogate for the actual patient. The data did not allow them to see patients "*as a whole.*" They stressed that to understand a patient clinically, they needed to "*look at the patient, talk to the patient, take care of the patient.*" Social workers shared that they had not met with this patient nor talked to their family. In our field evaluation, presentations of the synthetic patient cases were always followed by a long, awkward silence.

Clinicians also had wildly different readings into the same DST prognostics. We presented the same two synthetic cases with the same implant survival predictions to all participants. Interestingly, they generated wildly different reactions and interpretations of the cases. Some viewed the survival estimate as implying that an implant would not work. "*Gee... VAD is futile here.*" Others viewed the DST output as implying the patient should be immediately implanted, before things got worse. ' '*We still have a chance.*" That the data was based upon synthetic patient cases made any real discussion about the patient even more difficult. Instead, clinicians' discussions and questions depicted many unexpected intricacies in this notion of the "right" level of unremarkableness:

- *Is the Model Validated by Clinical Trials?* Clinicians commonly expressed a need to know more about the model's source and credibility. When they learned that the model presented has not been rigorously validated through clinical

trials and published in prestigious clinical journals, they suggested we were wasting their time. Physicians also desired a model that had been validated with data from their own hospital. "*It's better to be home-grown.*"

- *Are the Predictions Based on Clinicians' Best Efforts?* Physicians highlighted that the predictive models, regardless of how well they measure medical uncertainties, would never replace human, clinical decision-making. They viewed their own decision making as focused on managing and reducing uncertainties. "*If we think that we will be able to tell everybody what to do based on a model, we ignore the fact that we also have tools and mechanisms for dealing with the uncertainty that is inherent when putting VADs in patients.*" (Cardiologist)
- *Does DST Prediction Mean Causality or Correlation?* There was a sense that if the DST predictions were not based on causal factors, then the predictions should not be presented at all. Clinicians described differentiating correlation (predication) versus causality as a central part of their clinical decision making.
- *Are Data-Driven Prognostics Facts OR Predictions?* Clinicians frequently asked us to clarify whether DST prognostics are predictions that carry agency and subjectivity, or if predictions are facts rooted in historic data. We sensed they wanted to limit discussions to facts, including how heart failure has played out for the patient they were treating and the statistics from previous, similar cases.
- *Are the Predictions Individual Medicine OR Population Medicine?* Most clinicians seemed to find the notion of personalized predictions difficult to grasp. Some voiced strong concerns that using DST was the same as applying "*populational statistics*" to individual patient decision making.
- *What Does "Now" Mean in DST Predictions?* The DST visualized the patient outcome predictions. For example, it shows that the patient's post-implant life expectancy is 21 days if a VAD was implanted now, under the condition shown on the slides. Clinicians were confused by this notion of "now" because it was extremely unlikely that they would implant a patient on the same day as the decision meeting. "*Is that 21 days from today? If we are gonna lose the patient in 21 days [21 days following after implant], can we just wait?*"
- *DSTs Do Not Account For the X Factors* Clinicians said that the DST would only ever be one factor in their decision because of "*X factors*"; the many factors beyond a patient's condition that impacts the implant decision. One X factors they spoke of was O/E ratio (observed-to-expected mortality ratio). The O/E ratio is a rating that measures the surgeon and care teams' performance. Surgeons cared about keeping a high rating. They described the implant decision for high-risk patients as "*taking on new O/E ratio debts.*" This seemed to strongly influence whether they take on another high-risk patient. It seemed to depend strongly on how many patients had recently had poor outcomes.

Generalizability Beyond VAD. Our interviews with clinicians outside of VAD centers showed that multidisciplinary decision meetings take place across many clinical domains for some of their most aggressive interventions. They are also referred to as internal medicine panel meetings, tumor boards, or floor meetings (referring to meetings between critical and general care physicians). These meetings happen widely because for patients are very sick and are being considered for their last-option surgical intervention, their illness usually have involved multiple organs. Treating them requires physicians from multiple clinical domains. Multidisciplinary meetings therefore occurred naturally.

> *Esophageal cancer, COPD, diabetes, cystic fibrosis, LITERALLY everything in psychiatry, gastric bypass, end stage renal disease, hernia repair, syndromes like Down and Turner, any disease that requires management with meds with nasty side effects, and even emergency room situations to expedite processes. Any of the above diseases the approach has to be multidisciplinary almost by definition because they affect multiple systems and usually but not always the last option is a surgical intervention.* (Pediatric surgeon)

5.1.5 Project Summary

Clinical DSTs, despite compelling evidence of their effectiveness in labs, have mostly failed when moving out of labs and into healthcare practice. This project focused on designing DSTs not only as a functional utility but as an integrated experience. In doing so, we argue that clinical DSTs' effectiveness should be measured not only by prediction accuracy, but by effectiveness when situated within its social and physical context such as workplace culture and social structures.

The slide-builder DST design demonstrates one such design opportunity. The walk-up-and-use convention of current DSTs assume clinicians will know when they need help. Our design challenges this convention by proposing the notion of *Unremarkable AI*. Findings of our field evaluation suggested that unremarkable DSTs may more effectively fit into clinical practice if they can naturally augment the current activities of decision making, rather than pulling clinicians away from doing their routine. Taking lessons from prior HCI work, we should not only make AI more intelligent, but make them highly situated in people's routines. In doing so, AI can become part of the decision-making routines, part of the very glue of clinicians' everyday work.

DST's interaction should have a right level of unremarkableness, yet the information it provides should significantly impact care. While our DST was visually unremarkable, its very existence seems be, to an extent, transforming clinical decision making. It introduced predictions into a culture rooted in facts and statistical significance. Moreover, when predictive risk models were used officially to measure patient risk and clinician skills, clinicians' decision making became centered around these models. DSTs substantialize their performance pressure in decision making.

These observations forced us to take a step and ask: What is the preferred role for DST to play in clinical decision making? Where does a right level of unremarkableness lie? It is my hope that future work will continue to search for the right balance between DST *augmenting* decision-making in natural and intuitive ways and *transforming* the nature of clinical decision-making.

5.1.6 Broader Reflection on AI as Design Material

I also want to map the course of this project onto the framework of AI as a design material, and reflect on and examine how the findings of this project might inform the broader challenges of designing AI systems.

Conversation with AI capabilities. Almost all existing prognostic DSTs currently takes a context-less, prototypical form: It requires users to recognize that they need help, takes in a list of patient condition measures and produces an individualized prediction of patient trajectory [11]. In this project, we did not take this framing as given; Instead, we designed the DST as a situated experience in the rich social, physical, and procedural contexts of VAD decision making. As Figure 11 illustrates, we took AI as a design material, and reframed how it arrives and makes place in the clinical practice.

The framework (Figure 11) also revealed that our conversation with AI's capabilities was limited in this project. We took clinical DST's initial capability (mining patterns of medical history and intervention outcome) as a given. Our design and prototype were based on two simple conditions: DST predictions being in alignment or misalignment with clinicians' judgement. We did not consider the nuances of this capability (e.g., bias) or how it might shift when the DST gets deployed in the real world, or when encountered by various kinds of patients. In these aspects, our design solution can still subject to algorithm failures and biases. Therefore, in my following projects, I chose not to start with a preconceived notion of system capability, and instead actively work to engage with AI's capabilities at different levels.

Rapid Experience Prototyping AI Interactions in Critical Use Cases AI system's dependency on data and long development circle impose fundamental challenges to iterative UX design and evaluation. Making it even more difficult,

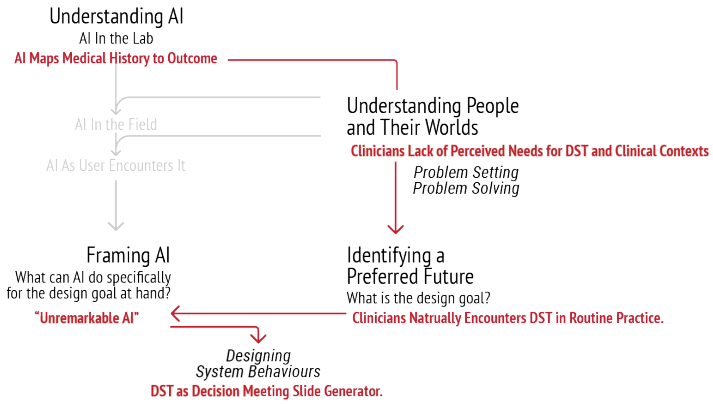

Fig. 11. Mapping the Designing a Clinical Decision Support Tool Project on the Framework of AI as Design Material

in this project, is the restricted access to the clinical environment. In this light, several tactics that seemed effective in this project might also help other designers in reducing the risks of one-shot AI design evaluation.

(1) Designing generalizable interactions: The work flows and social contexts in clinical practices are complex and highly divergent across hospitals. Therefore, generalizability is a necessity for many DST designs. This work took a step further than hospital-site generalizablility, designing a DST that can work for a class of structurally similar decisions (data-intensive, last-option surgical interventions). A DST's design and evaluation can become more productive than those dedicated to one specific clinical decision as well as specific DST models;
(2) Designing the evaluation methods for describing and unpacking the complex, subtle, and multifaceted nature of experience, rather than explicitly measuring it;
(3) Using prototypes rather than functioning AI systems. This tactic allowed us to probe various possible system outputs and to easily adjust our prototype to incorporate participant feedback.

Nonetheless, we encountered additional, seemingly-inevitable challenges of assessing DST's situated user experience. For example, whether a DST design has indeed achieved a right level of remarkableness was impossible to assess without real patient data and fully functioning ML systems. Clinicians need more than just synthetic patient cases to connect with their own decision making. We speculate that clinicians need to see one of their own patients' data to really assess the DST information design and to see what an actual prediction would look like. This means early DST prototypes will need actual patient data to assess their interactions in context and their impact on care. This is currently impossible in critical clinical cases due to ethics, policies and hospital regulations. It is my hope that future work will build upon the learnings of this project, and investigate more effective ways of experience prototyping AI systems.

5.2 Designing an Intelligent Text Editor

5.2.1 Background

From conversational agents at home to opinion mining on social media, intelligent language interactions are hitting the mainstream. Powering these systems are Natural Language Processing (NLP) technologies [2]. NLP makes it possible

[2]When we speak of NLP technologies, we broadly refer to any computer manipulation of natural language, ranging from simply counting word frequencies, to giving meaningful responses to human utterances [13]. Some examples of modern canonical NLP problems are information retrieval, machine translation, dialogue systems, and question answering.

to computationally comprehend, interpret, and even generate human language, and enables novel and otherwise impossible modes of interactions. But little is known about how user experience (UX) design practice might instigate novel products and services that use NLP.

This project aims to add intelligent language functionalities into a Word document editor. In collaboration with a group of NLP researchers, I started by attempting to rapidly experiment with many tentative NLP design ideas and broadly explore how NLP might improve the authoring experience. We wanted to use storyboards, UI wireframes, paper prototypes, and other simple, tangible instruments to sketch out early design ideas and probe users' reactions. But we soon encountered unexpected challenges. Common sketching tools and techniques deal with tangible interactions and are ineffective at abstracting the experience of language or a conversation. A number of technical aspects of language intelligence further complicate its UX design. For example, data-driven interactions vary across users, adapt to different contexts, and evolve over time, and it can be difficult for designers to envision such divergent courses of interactions or to visualize using traditional wireframes and prototypes.

These challenges led us to explore how to sketch NLP-powered user experiences, and what sketching actually means in the context of intelligent language interactions. We chose an Research through Design (RtD) approach: We first immersed ourselves in the concrete design problems of the project, and then offered an intentional accounting of the project to allow for objective reflections on procedural, pragmatic, and conceptual insights.

Under this research goal, our design goal was to improve individual users' writing *experience* with artificial language intelligence. This focus on *experience* means that we worked to understand how authors themselves want to be supported by machine intelligence and designed accordingly. Our focus was not on whether or how authors needed to write "better". Relatedly, we did not pre-identify the authors in need of assistance ("target user groups") either, because our very design task was to search for the authors in desperate need of assistance such that they would embrace the likely imperfect machine suggestions.

5.2.2 Challenges and Emergent Solutions of Sketching NLP

This process identified five challenges of sketching NLP, as well as the solutions emergent in our reflection in action. First, we conducted a field study of 18 participants to understand their needs and wants in writing. We invited them to record their screen for 40 minutes as they were writing one of their own documents. We then conducted an 1-hour interview. Participants walked us through their thought process in writing during the time of screen recording and discussed their unmet needs and wants toward writing assistance.

Next, we envisioned many intelligent functionalities that users would be likely to find valuable. In doing so, we encountered several challenges: (1) **How to stay abstract when envisioning and sketching language interactions?** Traditionally designers address this challenge by drawing storyboards. Storyboards capture the contexts and the holistic experiences of a macroscopic design idea, while dismissing its interface and interaction details. This doesn't work for language interactions; language as a form of interaction carries both the interface and the utility it manifests. We did not know how to sketch one without the other. Our solution to this challenge was a new format of wireframe, namely *the notebook* (Figure 12).

With the notebook framework, we started to ideate many writing assistance utilities that, based on our user study, users are likely to find useful. The first round of ideation generated 19 design ideas. To our great surprise, according to the NLP researchers, *none* of the ideas were promising from a technical feasibility perspective. These ideas "*need ten more years to make it happen,*" they said, only half-joking.

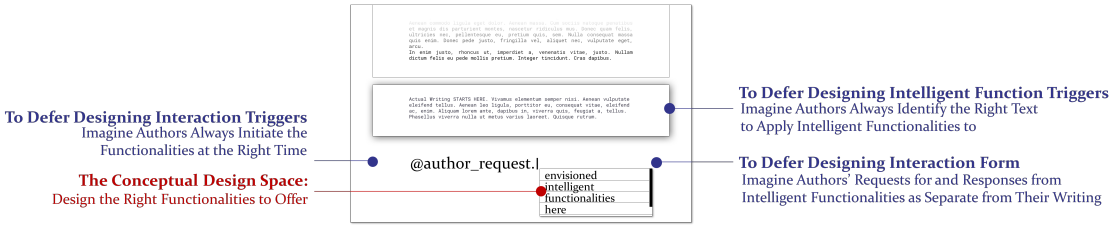

Fig. 12. "The notebooks" as a form of wireframe for sketching abstract, language-based interactions. It bounded us to focus on envisioning "the right thing to design" and deferred detailed interaction design tasks.

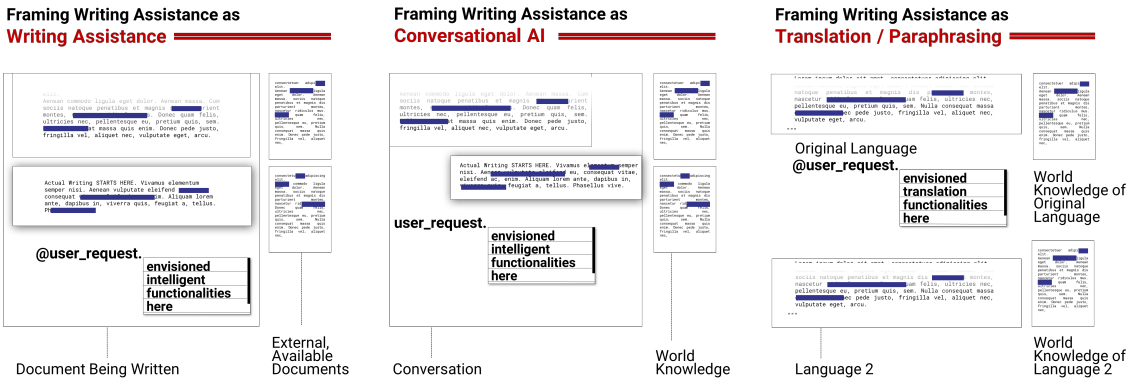

Fig. 13. Left: A developed version of the Notebook (Figure12), used as boundary object between HCI and NLP researchers. "Contexts" that can help inform intelligent function outputs are marked blue. Middle and Right: Reframing the problem of designing writing assistance as other canonical NLP technical problems. This expands our design space to the intersection between what authors want and what existing NLP capabilities can do.

Eager for more insights, we asked the NLP researchers: why are these designs technically unfeasible? Why does this functionality work in this research publication, but doesn't work for our design? What *is* feasible then? However, technical researchers were unable to answer these questions. They could not articulate NLP's technical limits with our abstract design ideas. "*It's difficult to say; It depends on data.*". "*The function you described is too abstract; I need to look at the data*".

In order to rapidly explore the design space, we could not afford to collect data before sketching. Data collection, prepossessing and exploration take up more than 80% of the total ML effort [109]. The question became: **how can we partner with NLP scientists without a text corpus at hand?**

After experimenting with numerous ways to explain our design ideas, we arrived at one boundary object [16] that effectively scaffolded our conversation with NLP researchers—that is, a more developed version of the notebook. This version of the notebook is projected on a Text Editor wireframe, resembling a user interface (Figure 13a). The notebook became a shared representation and a means of translation between the two worlds of UX and NLP. It scaffolded our discussions with NLP researchers for the rest of the project.

How can we understand NLP's technical capabilities and limits from an UX perspective? Realistically, to what extent can we push these limits to enable novel designs? A set of technical boundaries became clear to us after many discussions with NLP researchers, through negotiating with them and iterating on our design ideas. We describe these

boundaries via four measures of NLP's technical difficulty: text length, text classification-comprehension-generation, effort needed for labeling (for classification problems only), and likelihood to find training data that resemble the envisioned input/output pairs (Table 1). This set of measures enabled us to eventually find the intersection design space between what is valuable to users and what is technically feasible.

With some understanding of the technical limits, we sketched new design ideas that NLP researchers consider as implementable, or "*at least have a clear direction to work from*". However, we found our own design ideas rather unsatisfying. State-of-art NLP can assess or classify writings, but cannot easily pinpoint causes of the problems or generate suggestions on how to improve. This seemed a textbook recipe of a frustrating user experience. As a result, most of our design ideas are word or phrase-level alternations, "*many variations of auto-complete and auto-correct basically*".

How can we envision technically feasible NLP designs that have not been imagined before? How can we expand this narrow intersection between what is value to users and what can be built? We addressed these questions with a classic designerly approach: taking designing writing assistance as a wicked problem [19] and seeking reframings.

Our initial problem framing underlying the original notebook wireframe is that the writing assistance functionality comprehends and generates texts as the author writes. This framing is prone to technically challenging design solutions because authors are inherently better than algorithms at comprehending their unfinished writing and at predicting their unformed ideas. This is a wicked problem that cannot be accurately modeled. We reframed the relationship between authors and writing assistants as other canonical NLP problems, specifically human-AI conversations, information retrieval/search, and question answering. (Figure 13 illustrates how we abstracted users' writing into many text components, and then mapped them onto other NLP problems.) Each of these alternative framings exposed us to a new set of technical capabilities in a different NLP sub-domain.

Axes of NLP Capabilities	**What It Takes to Extend the Capabilities**
Text Length. Words are easier to computationally process than phrases, than sentences, than paragraphs and finally a document. Knowledge beyond the written texts (e.g., common sense) is the most difficult to process.	Escalating an intelligent functionality, for example, from word level to sentence level, requires building **new models** and "*there is no guarantee how well it will work.*"
Classification - Comprehension - Generation Assessing or classifying a piece of text is easier than comprehending it (i.e. pinpointing the problem in this text), than text generation.	Escalating intelligent functionality along this axes requires building a **new model** and collecting additional or new labeled **datasets** for building it.
[classification only] Efforts Needed to Label Training data. If it is *easy* and *fast* for humans to make an *agreed-upon* judgment of its class, curating a labeled dataset for this intelligent classifier is likely to be practical.	Time, effort and often financial costs. Medicated by the amount of labeled data needed.
Likelihood to find training data that resemble the envisioned input/output pairs. We cannot presume a model that performs well on a benchmark research corpus would naturally perform in other texts.	Transferring or generalizing an existing model to a different corpus requires building **new models** and "*there is no guarantee how well it will work.*"

Table 1. NLP Capabilities, Limits and What It Takes to Extend the Capabilities

5.2.3 Challenges and Emergent Solutions of Fast Prototyping NLP

After addressing the challenges of sketching, we proceeded with a small set of design ideas. For example, we envisioned an ask-your-reader function that compares a user's writing with their target venues', helping them to account for readers' likely expectations. We then turned to building a low-fidelity prototype to rapidly experiment on these ideas with users. We wanted to test the ideal behavior of our envisioned intelligent assistance with users to see if we were pursing the right design direction; We also wanted to probe users' reactions to a more realistic range of NLP-powered behaviors and errors to account for these reactions and expectations when improving on our design. But **how can we realistically simulate NLP's errors without spending months building fully-functioning systems?**

After much experimentation and interviews with technical researchers on various NLP sub-domains, we designed an alternative WoZ method to prototype our designs. For each NLP-powered interaction we envisioned, we designed a different hybrid of WoZ and off-the-shelf toolkits to best simulate the likely errors. The design of each hybrid mimics the likely architecture of its underlying NLP system. This method highlights that different intelligent features produce different kinds of errors, each of which can have different UX consequences. In order to better capture these consequences, we need to better orchestrate WoZ behaviors to simulate NLP behaviors.

For example, we simulated generative writing assistance functionalities with a multi-wizard simulator. When a user study participant request a piece of machine-generated text, multiple wizards and a meta classifier work in the background; each produces a response that excels at one aspect of the text generation. For example, one wizard produces a topically relevant response. The second wizard takes charge of the response fluency; the third focuses on the coherence between the generated text and the writers; the forth adds domain knowledge to the response, the fifth generates random words, and so on. The meta classifier assembles all wizard's responses into the final response returned to the user. We designed these wizards' roles based on common models of generative neural networks. We simulate different kinds/degrees of generative errors by tuning the weights that each wizard carry. As such, we could probe user study participants on their preference among various designs of a generative writing assistance as well as their error tolerance. These user study results can inform our future iterations of sketching and design refinement.

To summarize, we identified five challenges that are central to sketching NLP in our practice, and in the same process, three sketching instruments emerged and became useful to us.

- The notebooks. The notebooks are a set of wireframes that illustrate abstract language interaction design ideas. The notebooks ended up playing three important roles in our sketching process: They enabled us to externalize and communicate early-stage, abstract, language-based design ideas (challenges 1,4). It also served as a boundary object between designers and data scientists (challenge 2) and between designers and users (challenge 5), which enabled conversations among UX, design and NLP expertise;
- A set of NLP properties that are closely relevant to UX design, including "axes of NLP capabilities" and "what it takes to extend them". Understanding these properties helped us frame the design space within current technical limits (challenge 2);
- An alternative WoZ prototyping method. For each NLP-powered interaction, we designed a different hybrid of WoZ and off-the-shelf toolkits to best simulate the likely errors (challenge 5). This method shares the goals of traditional WoZ in enabling fast prototyping of NLP. In addition, our method highlights that different intelligent features produce different kinds of errors; each kind can have very different UX consequences. In order to better capture these consequences, we needed to better orchestrate WoZ to simulate NLP behaviors.

5.2.4 Reflecting on AI as Design Material

In this project, we started with a predetermined design goal (the preferred future in Figure 14), and actively engaged with the complexities both in terms of understanding NLP capabilities and those of author needs and writing contexts. Reflecting this wandering "conversation" between the two, I discuss two lessons learned.

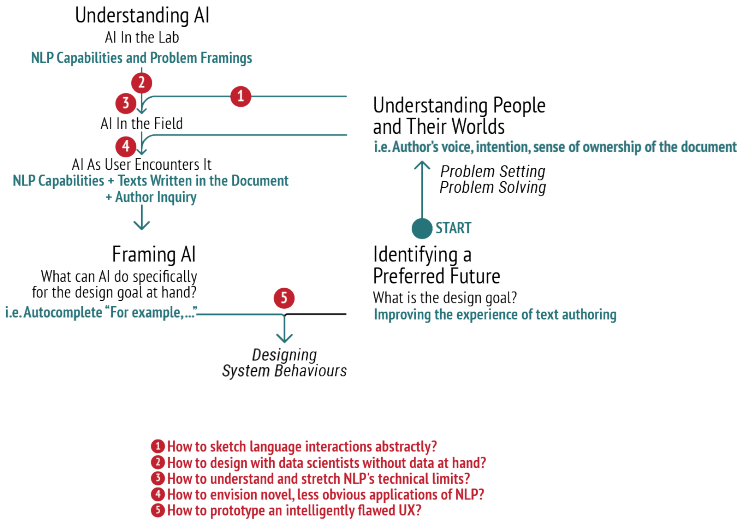

Fig. 14. Mapping the Designing Intelligent Text Editor Project on the Framework of AI as Design Material. Red circles marked the five challenges we identified.

The Importance of Abstraction. Abstraction is essential to any early design ideation, yet a missing perspective in NLP HCI literature. Most assumed that designers start designing by "*writing linear dialog examples*" [58]. WoZ studies often simulated NLP interactions with rule-based systems or crowd intelligence; Deliberations are lacking on whether these are effective abstractions of NLP system outputs.

This project draws attention to the importance of abstraction in sketching intelligent language-interactions. It was particularly important in the early design stages, the process of exploring many broad ideas before drafting concrete UIs or dialogues. For example, what kind of writing assistance do people even want? It seemed impossible to traverse such a design space by writing concrete dialogue examples, and it was not intuitive how to sketch dialogues abstractly.

NLP systems are difficult to design abstractly because both language-based interactions and intelligent interactions are difficult to abstract. On one hand, most sketching techniques and tools in designers' tool-belt, such as storyboards and wireframes, have evolved over the last two decades under the dominance of the graphical user interface, not directly applicable to language interactions. On the other hand, data-driven interactions are divergent, constantly evolving and sometimes make errors incomprehensible and difficult-to-anticipate. It could seem like only building a working NLP system can reveal its likely behaviors.

Upon reflection, all five challenges we encountered, and their solutions, involved some aspect of abstracting NLP interactions: Challenge 1, 2 and 4 dealt with abstracting interaction design ideas into different problem framings effective for design deliberation, communication and innovation respectively. The slightly different forms of the notebooks embodied these abstractions. In response to Challenge 3, Table 1 attempted to abstract NLP's technical capabilities and

limits, in order to bound the design space it enables. In response to Challenge 5, our prototyping methods dealt with challenges in abstracting the experiential qualities of NLP interactions, especially its errors.

In this light, supporting sketching language interactions *abstractly* is an important yet under-engaged issue for HCI/design research. The findings of this project revealed three aspects of abstraction, offering a starting place for this line of research.

(1) Abstracting language *interactions* as ways of framing its design problems
(2) Abstracting NLP *capabilities* to frame its design space realistically
(3) Abstracting NLP's *experiential qualities* to enable rapid UX prototyping

Designerly Understanding of NLP's Capabilities and Limits. Let me expand on the issue of "abstracting NLP capabilities". Like many other designers [30], we found understanding the capabilities and limits of intelligent technologies challenging. Our early sketches of writing assistance revealed a significant gap between how we wanted to support users ideally and what NLP can build realistically.

The findings of this project provide a glimpse into what a designerly understanding of NLP capabilities might look like. In this case study, three aspects of NLP were relevant to our design, as they emerge naturally in our design activities: (1) High-level understandings of NLP's capabilities and limits, which oriented our design ideation (Table 1 left); (2) NLP's capabilities given the available data and development resources, which informed our design deliberation and UX-gain-technical-investment negotiation (Table 1 right); (3) Each design's likely errors and other experiential qualities, which enabled rapid prototyping and helped us account for unexpected system behaviors.

5.3 Proposed Work: Designing NLP System for the Fanfiction Ecology

I have so far proposed an initial framework for designing AI. Through two RtD projects, I demonstrated the value of this framework in connecting the challenges encountered and learnings gained in concrete design situations to the broader research space of AI as a design material.

For the rest of my Ph.D., I plan to complete one more RtD project with the goal of further investigating ways to unfold designerly conversations with AI. Specifically, I propose to design a novel NLP product or service for the fan fiction (fanfic) writing-and-consumption ecology.

5.3.1 Background: Fanfiction and Its Communities

Fanfictions (fanfic) are works of creative writing that is transformative of existing media (e.g., novels, games, anime), borrowing setting, characters, or major plot elements to create a new story. Fanfictions are created within and for a fandom communities. The term "fan fiction" has been attested in print as early as 1939, was popularized via Star Trek fandom in the 1960s, and became even more popular and widespread since the Internet.

Fanfiction is a huge, rich and continuously growing corpus of literary works. A 2012 report estimated that fan fiction comprises one third of all content about books online [24]. For example, as of January 2019, there are almost 9 million Harry Potter fanfic stories on fanfiction.net and archiveofourown.org (AO3) alone. The most popular of these stories have received up to 35,000 reviews.

Underlying the fanfic works are many sizable, enthusiastic, self-motivated, and deeply engaging online communities. Today, much of fanfic communities' social activity takes place on Tumblr, with archival functions on platforms such as fanfiction.net and AO3. As of February 2019, AO3 alone has over 1.8 million users and 4.5 million fanfictions for more than 31,000 fandoms. An 2011 survey of 95,313 AO3 registered users showed that 49% of users used the site

at least once a day. Half of them spent more than 5.5 hours a week on AO3. The survey also showed that 90.3% of respondents self identified female; 54% self identified as a gender, sexual or romantic minority. 78% of them are white. Hereby fanfic communities have fascinated media studies and gender studies for decades as sites of studying parasocial relationships [50], online social norms [34], teen identity development [86], pop culture [88], gender identity [8], queer culture and more. These studies typically took an ethnographic approach, studying no more than 12 fanfic writers or readers a time.

In contrast to the size, intensity and complexity of the fanfic ecology, HCI research efforts can be seen as sparse on mining the fanfic corpus or understanding/supporting it communities. The few HCI and CSCW publications on this topic have so far focused on understanding this community via the lenses of language learning [?], copyright [36], and feminist value [37]. At CSCW'18, researchers in an panel discussion titled "Online Fandom: Boldly Going Where Few CSCW Researchers Have Gone Before" discussed vast opportunity in this area, and called for more work from a wider range of perspectives [31].

5.3.2 Study Plan and Timeline

The project has two stages. First, I plan to widely explore the design space centered around the fanfic ecology. I will begin by interviewing participants of the fanfic ecology and at the same time, exploring technical capabilities in this space with my technical collaborators. I will then synthesize these two body of insights, and work to identify the right thing(s) to make via matchmaking. Then in stage 2, I will proceed with one design idea and prototype it.

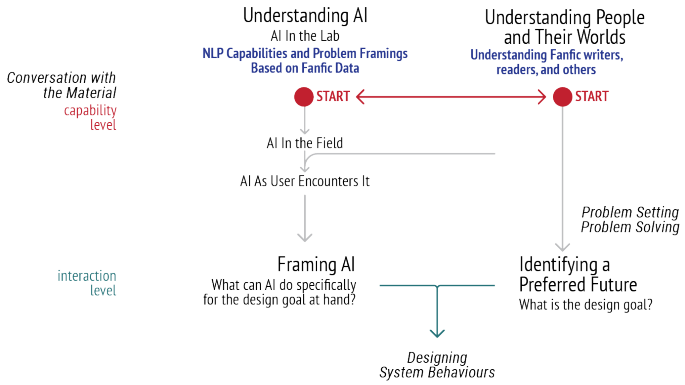

Fig. 15. Mapping the Fanfiction Project Study Plan on the Framework of AI as Design Material.

Figure 15 illustrates how this two-stage process will help extend existing understanding of designing AI. In the first stage, I will focus on the conversations between the technical capabilities of AI and different lenses into the fanfic ecology. In doing so, I hope to advance existing understanding of "match-making" as a method for transferring AI technologies to design innovations.

Match-making is a design method that designers start with technology and search for valuable new things to make [14]. We speculated that this process would benefit AI's design innovation, and the studies so far described in this proposal echoed this speculation. Match-making can be seen as a key touch point between UX and technical AI expertise, an important process where the designers' conversation between the material of AI happens. Experienced designers in the

industry undertook matchmaking through constant exchange with technical AI experts and extended this process in similar ways as building lean startup (Section 3.2). The two RtD projects offered additional points of reference: In the healthcare project (Section 5.1), match-making was bounded by the limited amount and affordance of clinical datasets, therefore performed mostly during field work; We took DST's lab capability as a given and sought for appropriate time and place in the clinical practice where a DST might make its place and add valuable. In the Text Editor project (Section 5.2), we took a pre-identified UX goal – improving authors' writing experience – as a starting point, and performed match-making by identifying many user needs and then eliciting ones within the limits of state-of-art NLP technology. This approach to matchmaking was successful in that it effectively enabled us to rapidly traverse the design space and to identify viable opportunities. However, we were unable to more concretely deliberate on or evaluate the "matches" made.

The fanfiction ecology provides an opportune opportunity for me to extend this line of research inquiry. It offers a hopeful ground for matchmaking, because it encompasses both a large amount of accessible data and a wide variety of human perspectives and rich social and cultural contexts.

I plan to complete the interviews, design ideation and match-making before 2019 fall, and to implement the prototype and conduct a second user study with it over the 2019 fall semester. I hope to defend my thesis before the end of 2020 Spring semester.

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